Major Project Report

On

Analyzing the Evolution of YOLO Models with Robustness Assessment under Standard Noise Conditions

Submitted in the partial fulfillment for the award of the
Degree of Bachelor of Technology

In

Electronics and Communication Engineering

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DECLARATION

We the undersigned solemnly declare that this report of the major project work, entitled **"Comparative Analysis of YOLO Architectures (v8–v26) for Human Detection Robustness in Stochastic Noise Environments"** is carried out during the course of our study during VIII semester under the guidance of **Mr. Jitendra Bhardwaj**, Assistant Professor, Department of Electronics and Communication Engineering, School of Studies in Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.). We further declare that this mini project work is presented for the partial fulfillment of the requirement of degree of Bachelor of Technology in Electronics & Communication Engineering, School of Studies in Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur (C.G.).

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This major project report entitled "**Comparative Analysis of YOLO Architectures (v8–v26) for Human Detection Robustness in Stochastic Noise Environments**" by **Anand Kumar Rai**, **Mohammad Shahnawaz** and **Vimlesh Yadav** is approved for the partial fulfillment of the requirement of the degree of Bachelor of Technology in Electronics and Communication Engineering.

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Date: 07/05/2026

Place: Bilaspur

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SCHOOL OF STUDIES IN ENGINEERING AND TECHNOLOGY

GURU GHASIDAS VISHWAVIDYALAYA, BILASPUR (C.G.)

(A Central University established by the Central University Act 2009 No. 25 of 2009)

CERTIFICATE

It is certified that the major project entitled **"Comparative Analysis of YOLO Architectures (v8–v26) for Human Detection Robustness in Stochastic Noise Environments"** submitted by **Anand Kumar Rai**, **Mohammad Shahnawaz** and **Vimlesh Yadav** in partial fulfillment of the requirements of the award of the degree of Bachelor of Technology in Electronics and Communication Engineering, School of studies in Engineering and Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur, is carried out by them in the Department of Electronics and Communication Engineering during session 2025-26 under supervision and guidance of **Mr. Jitendra Bhardwaj**, Assistant Professor, Department of Electronics & Communication Engineering, School of Studies in Engineering & Technology, Guru Ghasidas Vishwavidyalaya, Bilaspur CG.

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Abstract

This study evaluates the operational resilience of modern real-time object detection architectures specifically the YOLO (You Only Look Once) family ranging from version 8 to the experimental version 26 under diverse environmental noise conditions. Using a subset of the COCO dataset focused on human detection we systematically augmented three types of stochastic noise: Gaussian, Salt & Pepper, and Poisson across six noise intensity levels. We evaluated performance using Mean Average Precision (mAP), F1-Score, Inference Latency and robustness retention index. Our results reveal a significant "performance cliff" for nano-scale models at high Gaussian variance, while Poisson noise exhibited the least impact on detection count. Notably, newer iterations like YOLO26n and YOLOv11n demonstrated superior feature extraction stability, retaining over 90% of baseline mAP in moderate noise while maintaining sub-200ms latencies. This research provides a quantitative benchmark for selecting object detection models in noise-heavy applications such as automated surveillance and low-light search-and-rescue.

Keywords: object detection, edge deployment, noise robustness, YOLOv8, YOLOv9, YOLOv10, YOLO11, YOLO26, Gaussian noise, salt-and-pepper noise, lightweight deep learning

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1.0 Introduction

1.1. Object Detection

Object detection is a fundamental and impactful field in computer vision. Its goal is to identify and precisely locate objects within an image by providing a bounding box and class label for each instance. This technology serves as the core component for a vast range of real-world applications, including autonomous driving, smart surveillance, industrial automation, medical diagnosis (e.g., tumor detection), and robotics.

Over the past decade, deep learning, specifically Convolutional Neural Networks (CNNs), has led to remarkable progress. Among the various architectures, the YOLO family of models has emerged as a dominant force. YOLO re-frames object detection as a single regression problem, processing the entire image in one forward pass. This design, refined through successive versions like YOLOv8, offers an unparalleled combination of high-speed inference and strong accuracy, making it the preferred choice for real-time systems.

1.2. The Challenge of Real-World Noise

While recent YOLO variants, from YOLOv8 to YOLOv26, achieve strong performance on clean benchmark datasets such as COCO dataset, real-world deployment conditions are often far less controlled. In real world human detection scenarios, input images are frequently affected by different forms of noise and signal degradation which are particularly important for lightweight models.

These disturbances arise from multiple sources. Sensor thermal variance, caused by electronic interference and heat within imaging devices, introduces random fluctuations that can be approximated as Gaussian noise. Impulse interference, often occurring during data transmission from edge devices such as drones or remote surveillance systems, leads to sudden pixel corruption and is commonly represented as Salt-and-Pepper noise. Additionally, challenging lighting conditions, particularly in low-light environments, result in shot noise due to the statistical nature of photon arrival, which can be modeled using Poisson distributions.

Such noise degrades the quality of visual information by altering pixel intensities and disrupting fine spatial details, including edges and textures that are essential for reliable feature extraction. Within the YOLO backbone, this can lead to feature corruption, where convolutional filters respond to noise-induced artifacts instead of meaningful patterns like human silhouettes, ultimately reducing detection reliability.

In domains such as autonomous navigation, pedestrian protection, and security monitoring which are safety-critical, consistent model performance is very essential. A system that shows a sharp decline in accuracy under noisy conditions can pose serious risks in real-world deployment. This work investigates

whether recent architectural improvements in newer YOLO variants, including the C3k2 blocks in YOLOv11 and the compression strategies introduced in YOLOv26, contribute to better robustness retention compared to earlier versions.

1.3. Project Objectives

The primary objective of this study is to systematically benchmark and compare the robustness of key YOLO models with the focus on the small and nano models for devices with low computational capability. The versions v8, v9, v10, v11, and v26. The focus is to evaluate how advancements in feature extraction and spatial modeling affect a model's ability to sustain detection performance under degraded visual conditions.

To achieve this, the study analyzes model performance under progressively increasing levels of three types of stochastic noises. Gaussian noise is used to represent thermal and electronic variations in imaging sensors. Salt-and-Pepper noise simulates impulse disturbances and transmission errors commonly observed in edge-device communication. Poisson noise is used to model signal-dependent variations typical of low-light environments. Through this evaluation, the study aims to quantify how well each model maintains detection accuracy as noise severity increases.

2.0 Literature Review

This section provides the necessary background on the object detection model utilized in this study and reviews the existing literature on the robustness of deep learning models to common image noise, thereby establishing the foundation and justification for the present work.

Model	Year	Params (n)	Key Architectural evolution	Tasks
YOLOv8n	2023	3.2 M	Decoupled detection head for improved	Mask-based instance segmentation, 2D Skeleton keypoint estimation.
YOLOv9n	2024	2.0 M	PGI and GELAN backbone to improve feature aggregation	Lossless-feature detection (ideal for medical/industrial imaging).
YOLOv10n	2024	2.3 M	End-to-End NMS free architecture	High-speed mask generation, Low-latency keypoint tracking.
YOLOv11n	2024	2.6 M	Optimized backbone and feature pyramid	Small-object specialized detection, Oriented Bounding Box (OBB) for angled objects.
YOLO26	2026	2.4 M	Transformer style attention block	Edge-optimized segmentation, Real-time Rotated Object and Pose detection.

Table 1: Architectural comparison of YOLO models.

2.1. The YOLO Model Architecture

2.1.1. YOLOv8: Anchor-free design

YOLOv8 introduces an anchor-free design that changes how bounding boxes are predicted, making the model more adaptable under noisy visual conditions [1]. By removing predefined anchor boxes, the model directly estimates object locations, which improves flexibility and reduces sensitivity to distortions caused by pixel-level corruption.

The YOLOv8 uses the C2f module in the backbone to extract features from the input image. This module splits the feature flow into two paths: one path processes the features through multiple transformations, while the other preserves the original information through a shortcut connection. This design helps maintain stable feature representations when noise is present. For example, under Gaussian noise, even if one path is affected, the shortcut path retains important spatial details, preventing the loss of key object structures such as human shapes.

In the neck, YOLOv8 employs a decoupled Path Aggregation Network (PANet) [] to combine information

from different feature levels. This structure allows both low-level details (such as edges and shapes) and high-level context (such as object identity) to interact.

This is particularly useful in the presence of Salt-and-Pepper noise, which introduces random bright and dark pixels that can resemble false edges. By integrating information across levels, the model can better distinguish real object boundaries from noise artifacts.

YOLOv8 uses a decoupled head that separates object classification from bounding box regression. This separation allows the model to handle recognition and localization independently. As a result, under Poisson noise conditions, which are common in low-light environments, the model often maintains correct object classification even if the predicted bounding boxes are slightly imprecise. This design contributes to more reliable detection performance in challenging visual settings

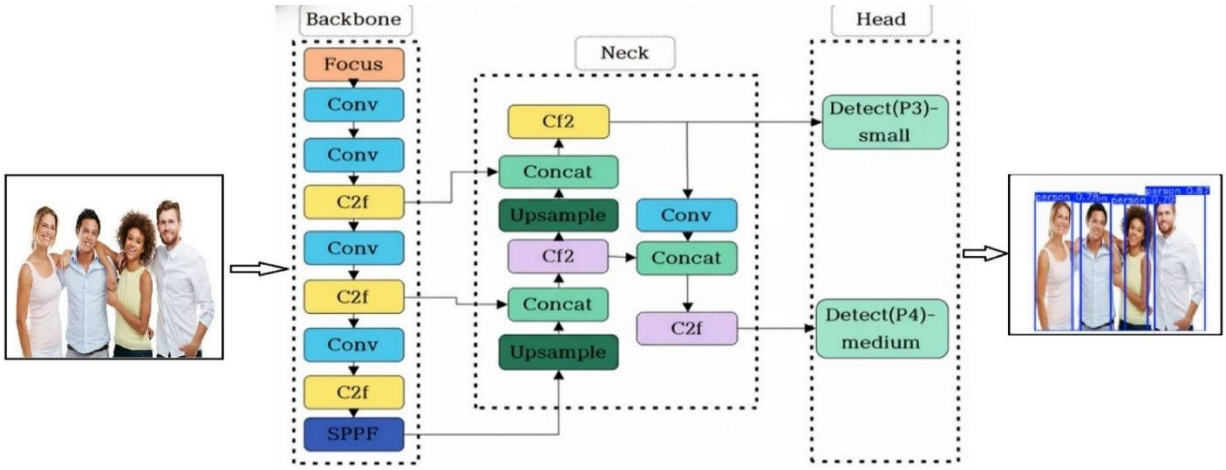


Figure 1: Architecture of YOLOv8

2.1.2. YOLOv9: Programmable Gradient Information (PGI)

YOLOv9 introduces the **Generalized Efficient Layer Aggregation Network (GELAN)** as a core architectural improvement to address information loss in deep neural networks [2]. In conventional designs, repeated transformations across layers can gradually degrade useful feature representations, especially in lightweight models with limited parameters.

GELAN mitigates this issue by enhancing how features are aggregated and reused across the network. Instead of relying on simple sequential propagation, promoting richer multi-path information flow. This leads to improved parameter efficiency which is critical for the lightweight devices.

In addition to GELAN, YOLOv9 introduces **Programmable Gradient Information (PGI)**,[2] a mechanism designed to improve the quality and reliability of gradient propagation during training. In deep networks, gradients can become weak or noisy as they pass through many layers, which negatively affects learning. PGI addresses this by explicitly preserving essential information and guiding how gradients are transmitted across the network. This ensures that important features learned in earlier stages are not lost

and continue to influence later layers effectively. By stabilizing the training process and maintaining informative gradients, PGI enhances convergence and leads to more robust feature learning.

Together, GELAN and PGI enable YOLOv9 to maintain strong detection performance [2] [3], particularly in challenging conditions, without compromising efficiency. As a result, YOLOv9 achieves a better balance between speed, accuracy, and model compactness, making it particularly suitable for real-time and compute-constrained devices [3].

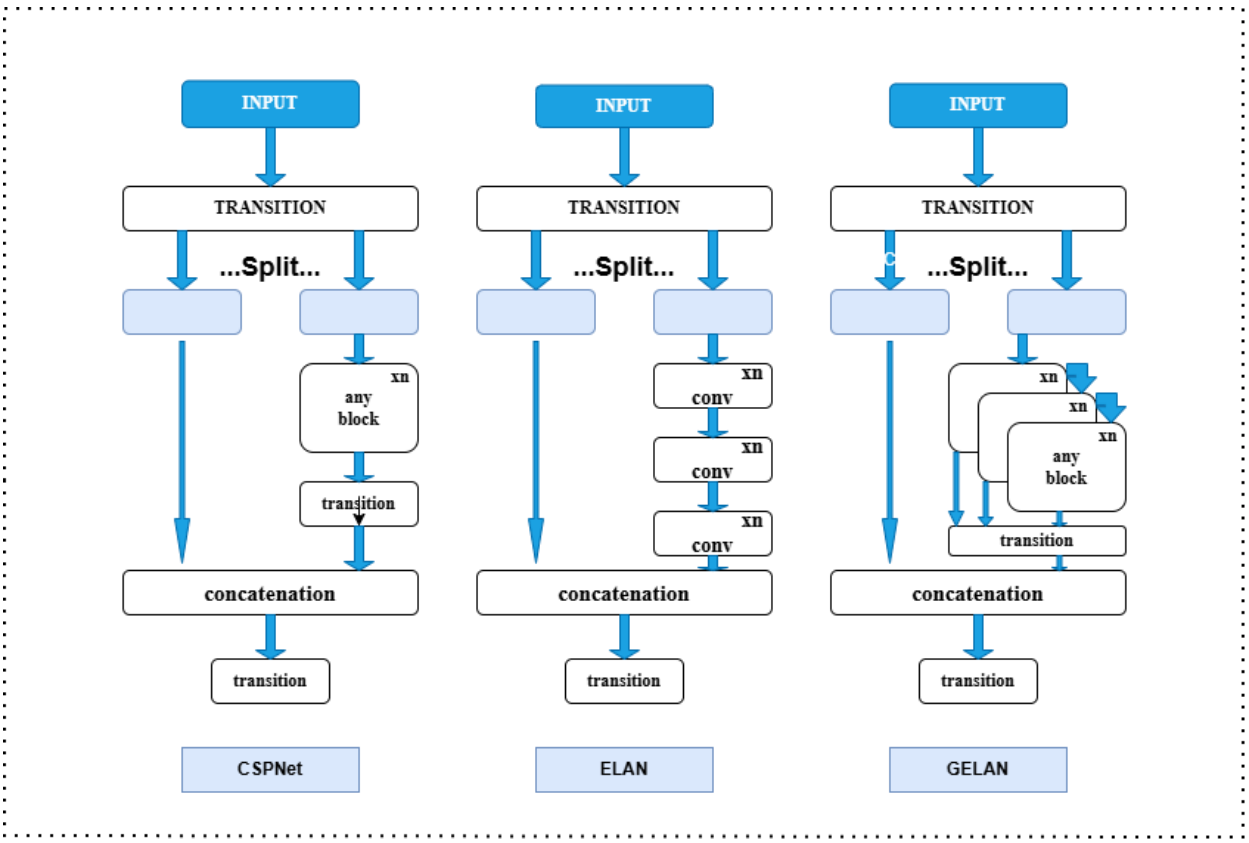


Figure 2: Architecture of YOLOv9

2.1.3. YOLOv10: NMS-Free Efficiency

YOLOv10 is designed to remove the post-processing dependency on Non-Maximum Suppression (NMS) that enables a fully end-to-end detection pipeline [3]. In earlier models, multiple overlapping predictions are generated for each object and later filtered using NMS, which adds latency and can become unreliable under noisy conditions. YOLOv10 produces a single, high-quality prediction per object directly which eliminates redundant outputs and reduces inference overhead.

In the backbone, YOLOv10 adopts a spatial–channel optimized design to improve efficiency while preserving feature representation quality [3]. It includes the selective use of depth-wise separable convolutions in place of standard convolutions to reduce computational cost and parameter redundancy. This results in ,stable feature extraction and consistent performance across different types of noise.

In the neck and head, YOLOv10 introduces a consistent dual assignment strategy. During training, the

model benefits from richer supervision through multiple candidate assignments, while inference enforces a one-to-one matching between objects and predictions. This ensures that each object is associated with a single bounding box, removing the need for NMS. As a result, the model avoids the computational cost of processing and filtering large numbers of overlapping predictions, leading to significantly lower inference latency. This design makes YOLOv10 particularly well-suited for real-time applications such as surveillance and autonomous systems, where both speed and robustness are critical [4].

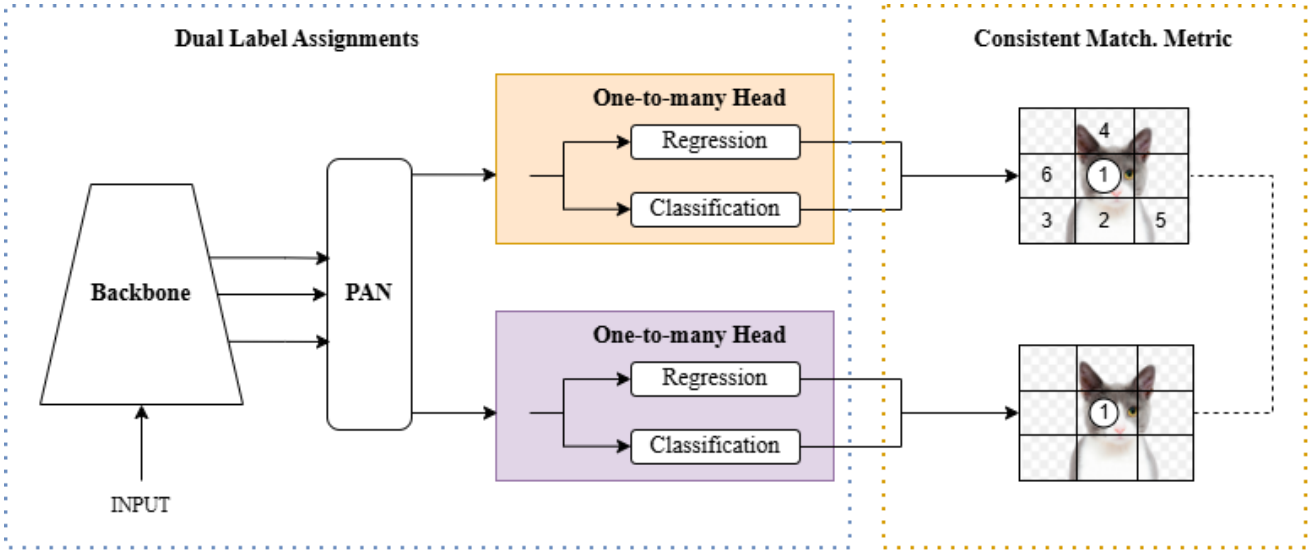


Figure 3: Architecture of YOLOv10

2.1.4. YOLOv11: Spatial Attention and C3k2 Blocks

YOLOv11 advances the architecture toward a more attention-focused design, with the goal of improving robustness in complex and cluttered environments. By emphasizing relevant regions and suppressing background noise, the model is better equipped to maintain detection performance under challenging visual conditions.

YOLOv11 introduces C3k2 blocks as an evolution of earlier C3-based modules in the backbone layer [4]. These blocks utilize smaller and more efficient kernel groupings, allowing the network to achieve deeper feature extraction without a significant increase in computational cost. The use of smaller kernels improves the model's ability to capture fine-grained spatial patterns. This is particularly beneficial under Poisson noise, commonly observed in low-light scenarios, where pixel intensity fluctuates unpredictably. The refined kernel structure helps the model focus on consistent structural cues, such as the texture and shape of a human figure, rather than being misled by grainy noise artifacts.

It incorporates a spatial attention mechanism to enhance feature selection, applying a learned weighting mask over feature maps and increases the influence of important regions while suppressing less relevant or noisy areas. As a result, the model effectively prioritizes meaningful object regions and reduces the impact of background clutter. This selective focus contributes to improve the perform enabling YOLOv11 to maintain stronger detection performance compared to earlier versions such as YOLOv8, particularly in

noisy or visually degraded environments.

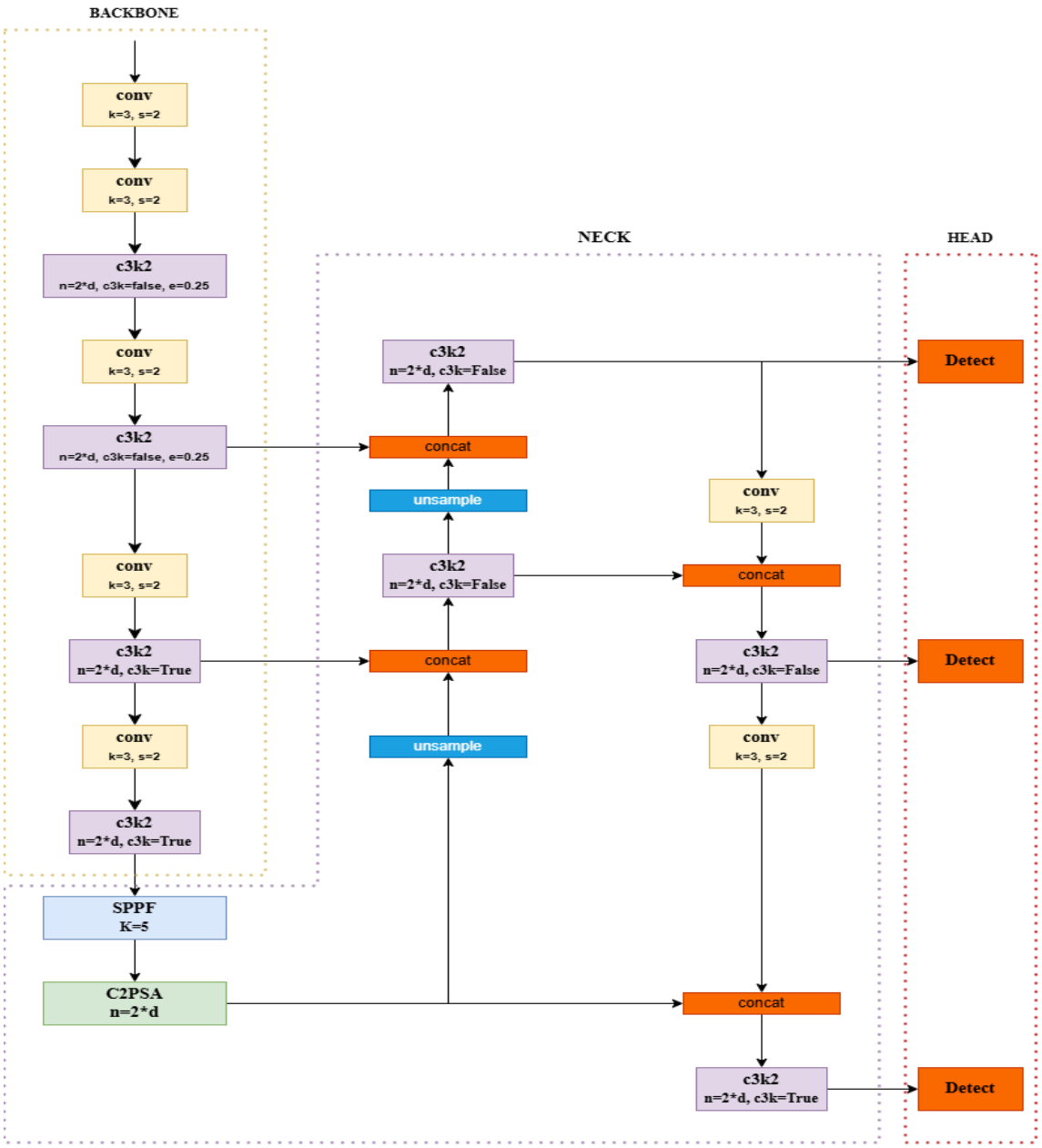


Figure 4: Architecture of YOLO11

2.1.5. YOLO26: Multi-Scale Contextual Fusion

The **YOLO26** model is designed as an edge-oriented object detection framework, emphasizing computational efficiency and deployment robustness on resource-constrained devices. Its architecture eliminates both **Non-Maximum Suppression (NMS)** and **Distribution Focal Loss (DFL)**, thus simplifying the inference pipeline and improving hardware compatibility. This redesign enables true end-to-end inference, removing post-processing overhead and reducing latency in real-time applications.

A key contribution of YOLO26 is the introduction of **Progressive Loss Balancing (ProgLoss)** which is a dynamic training strategy that adaptively schedules the relative weights of classification, localization, and auxiliary loss components. This mechanism prevents imbalance caused by easy negatives or dominant

large-object instances during later training stages, resulting in smoother convergence and improved optimization stability.

Using **Small-Target-Aware Label Assignment (STAL)**, which modifies assignment priors and spatial tolerance criteria to better capture small, occluded, and low-contrast objects. This enhancement significantly improves recall in edge-centric scenarios such as UAV imagery, smart surveillance systems, and mobile robotics.

It also integrates the **MuSGD optimizer** that is inspired by optimization strategies used in large languagemodels reducing the inference cost, making it suitable for edge devices

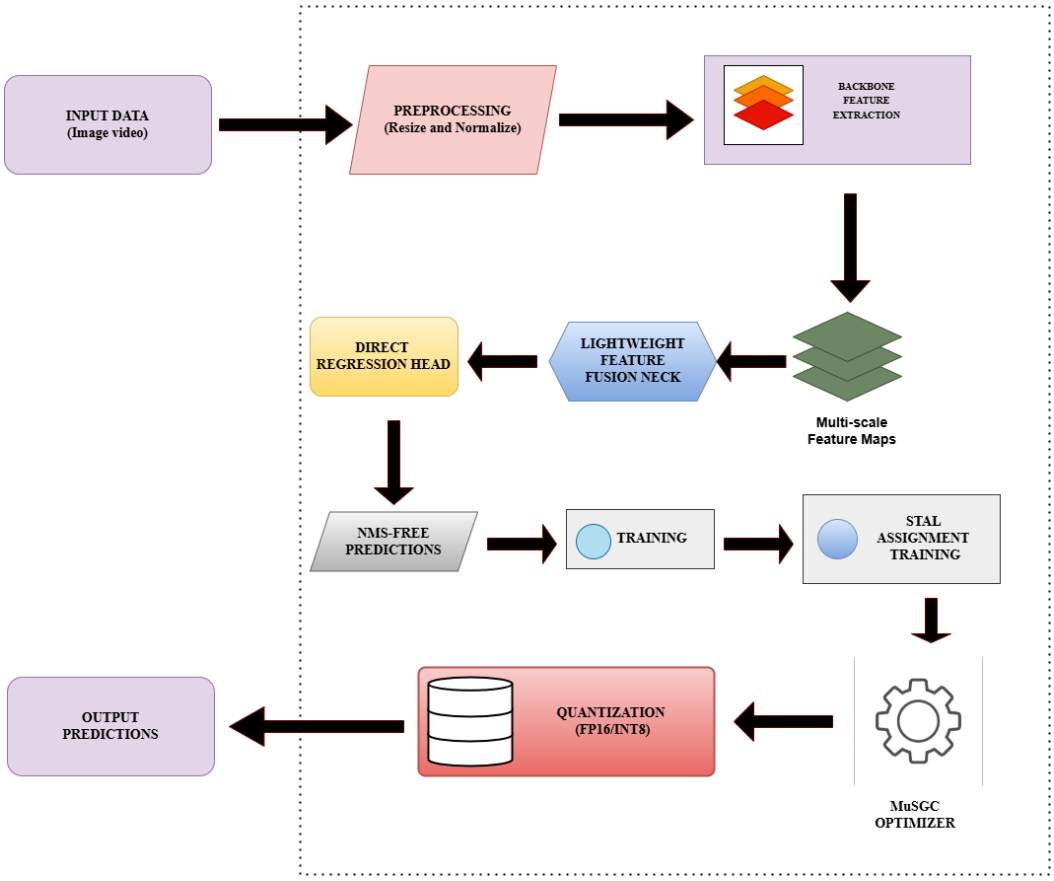


Figure 5: Architecture of YOLO26

Detailed Architectural Comparison

Property	YOLOv8n	YOLOv9t	YOLOv10n	YOLOv11n	YOLO26n
Core backbone block	C2f	GELAN	CSP + large kernel	C3k2 + C2PSA	
Neck	PANet	PANet	PANet	PANet	PANet
Detection head	Decoupled anchor-free	Decoupled anchor-free		Decoupled anchor-free	Decoupled DFL-free NMS-free
Attention module	None	None	Partial self-attention	C2PSA (spatial)	CNN+ Transformer style

Gradient strategy	Standard	PGI	Dual assignment learning	Hybrid task-aware	PGI
Anchor type	Anchor free	Anchor free	Anchor free	Anchor free	Anchor free

Table 2: Architectural comparison of changes in blocks of successive YOLO models.

2.2 Sensitivity of Deep Learning Models to Stochastic Noise

Despite the Although the YOLO family (v8–v26) achieves strong performance on standard datasets such as COCO, its reliability in real-world environments remains a significant challenge. In practical scenarios, images are often affected by various forms of natural noise caused by hardware limitations, low-light conditions, or transmission errors. These factors introduce randomness into the input data, which can degrade model performance.

This sensitivity arises from how these models learn and interpret visual features. Early stages of processing focus on identifying simple patterns such as edges and textures [11]. When Gaussian noise is present, it adds random variations across pixels, making these patterns harder to distinguish. As a result, the model may focus on irrelevant details rather than meaningful shapes, leading to incorrect detections.

Salt-and-Pepper noise has a different effect, as it randomly replaces pixel values with extreme intensities. This disrupts the consistency of the image, particularly affecting how the model combines information from different scales. Such disruption can cause the model to misinterpret random noise as actual objects, increasing the likelihood of false detections.

In low-light conditions, Poisson noise introduces uncertainty that depends on the brightness of the image. Darker regions become more unstable, making it difficult for the model to confidently recognize objects. While this type of noise may not completely obscure features, it reduces the reliability of predictions, especially for faint or partially visible objects.

Most prior research has focused on intentional attacks designed to mislead models. However, this work emphasizes natural noise, which is more commonly encountered in real-world applications. A key aspect of this study is identifying the point at which increasing noise causes a sudden drop in performance. This threshold helps evaluate how well different model versions handle real-world uncertainty and whether newer designs are more resilient under such conditions.

2.3.1. Prior Studies on Noise Robustness

Recent academic inquiries have extensively analyzed the vulnerability of deep learning architectures to stochastic and adversarial perturbations. Foundational work by Goodfellow et al. established that even infinitesimal, pixel-level alterations—initially conceptualized as adversarial attacks—can catastrophically disrupt neural network inference.

Subsequent research transitioned toward "natural" or non-adversarial noise models. Investigations into Gaussian noise demonstrated that high-variance perturbations severely degrade the feature extraction layers of Convolutional Neural Networks (CNNs) by obscuring high-frequency gradients [7]. Similarly, studies on Salt & Pepper (impulse) noise revealed a total collapse of structural cues, as these intensities (0 or 255) directly corrupt the pixel-level spatial consistency required for object localization.

In the specific domain of real-time object detection, the YOLO family has been benchmarked under environmental distortions such as motion blur, fog, and brightness variance. These studies consistently indicate a "Performance Cliff": a threshold where the model maintains high Mean Average Precision (mAP) under mild distortion but experiences an exponential decline as entropy increases.

2.5. Identified Research Gap

Despite the maturity of the YOLO framework, significant research gaps remain, which this project addresses through a multi-generational evaluation:

1. **Under-Exploration of Poisson (Shot) Noise:** While Gaussian and Salt & Pepper models are well-documented, there is a lack of extensive benchmarking for **Poisson noise**. This is a critical oversight, as Poisson noise is the primary limiting factor in photon-limited scenarios such as night-vision surveillance and low-light search-and-rescue. This project fills this gap by quantifying how models like **YOLOv11** handle signal-dependent grain.
2. **Lack of Generational Comparative Analysis:** Most existing literature evaluates a single iteration (e.g., YOLOv5 or v8). There is a significant lack of data comparing how architectural shifts—such as the **C2f** blocks of v8, the GELAN of v9, and the NMS-free heads of v10 differ in their "Noise Retention" capabilities. This study provides the missing link.
3. **Absence of Integrated Efficiency-Robustness Metrics:** Few studies simultaneously evaluate qualitative visual failure (what the model "sees") alongside a dual quantitative analysis of **Inference Latency (ms)** and **Robustness Retention Index (RRI)**. By plotting the **Efficiency-Robustness Pareto Frontier**, this project provides a decision-making framework for selecting models based on real-world environmental stress rather than just "clean" benchmark scores.

3.0 Methodology

The workflow presents a structured pipeline for evaluating object detection models under noisy conditions. It begins with data acquisition using 270 COCO images of the person class, resized to 640×640 pixels. Noise-based augmentation is then applied by adding Gaussian, Salt & Pepper, and Poisson noise to create distorted datasets. After ensuring annotation consistency through harmonization, multiple YOLO models (YOLOv8 to YOLO26 variants) are selected and configured for evaluation. Finally, model performance is assessed using metrics such as mAP, F1-score, and latency on noisy datasets, followed by result visualization to analyze robustness.

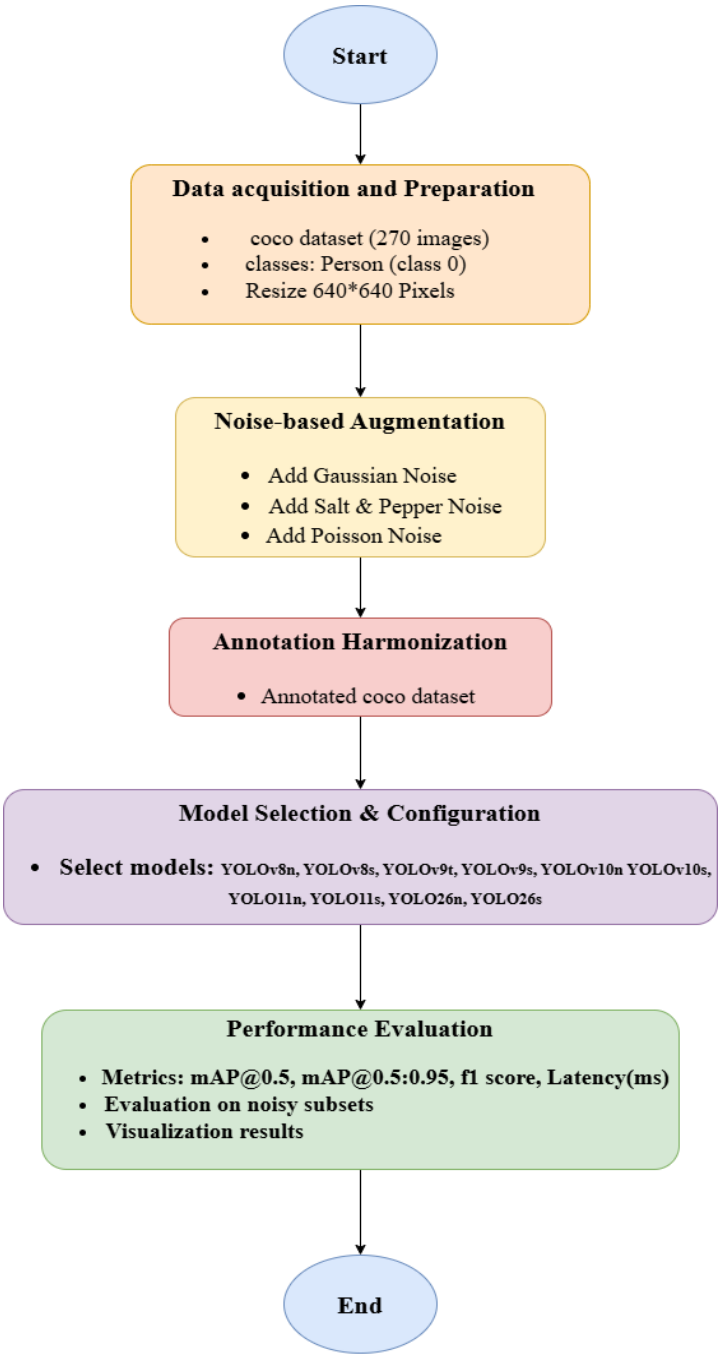


Figure 6: The complete workflow of the proposed methodology.

3.1. Dataset and Ground Truth Calibration

3.1.1. Strategic Selection from the COCO Validation Subset

The integrity of a robustness study depends on the use of a standardized, high-variance dataset that reflects real-world environmental complexity. For this project, we moved beyond anecdotal image testing to utilize a specialized validation subset from the **Microsoft COCO (Common Objects in Context) Dataset**. COCO is widely regarded as the definitive benchmark for training and evaluating real-time object detectors because it captures objects in "natural" contexts—meaning they are frequently partially occluded, viewed from difficult angles, or placed against high-entropy backgrounds. While COCO contains eighty distinct object categories, this study specifically isolates the **"Person" class**. Human detection is the primary safety-critical task in computer vision, serving as the foundational technology for pedestrian avoidance in autonomous vehicles, smart surveillance, and low-light search-and-rescue operations.

3.1.2. Establishing a Complexity Gradient

By extracting representative frames from the COCO validation set, we established a **"Complexity Gradient"** to evaluate the YOLO family (v8 through v26). This gradient ensures that our noise-injection tests are performed on images with varying levels of spatial and environmental difficulty. For instance, images with high background entropy (clutter) are used to test whether the model's **Backbone** (e.g., the **GELAN** architecture in YOLOv9) can distinguish a human silhouette from complex surroundings when **Gaussian noise** is added. Simultaneously, dense and crowded frames featuring up to 24 individuals are used to test the **Localization Head**. These high-density scenarios highlight the efficiency of the **NMS-Free heads** in **YOLOv10**, which must predict distinct bounding boxes for overlapping individuals despite the presence of destructive **Salt & Pepper** noise.

3.1.3. Evaluation of Low-Contrast and Environmental Variance

Furthermore, the dataset includes specific COCO frames selected for their low-contrast lighting and indoor shadows. These scenarios are critical for evaluating **Poisson (Shot Noise)** resilience, as they simulate the photon-limited conditions typical of night-vision imaging. This project benchmarks whether the **Spatial Attention** mechanisms integrated into **YOLOv11** can successfully "look through" grain and sensor artifacts to find consistent human features. By utilizing these standardized COCO frames, we ensure that our results are not biased by arbitrary data selection, but rather reflect the "Natural Robustness" of the models against the very data they were originally trained to identify.

3.1.4. Manual Calibration and the "Gold Standard" Ground Truth

While COCO provides professional-grade annotations, this study employs a rigorous **Manual Calibration Protocol** to verify the **Ground Truth (GT)** for our specific test subset. Every "Person" in the selected frames was cross-referenced against the master COCO annotations to ensure 100% mathematical accuracy. This manual verification ensures that our **Intersection over Union (IoU)** and **Recall** calculations are based on an immutable "Gold Standard." In cases of heavy occlusion, where people are only partially visible, our GT preserves the exact coordinates of the visible fragments. This allows us to measure the precise "Breaking Point" or "Performance Cliff" where environmental noise makes a partially obscured person completely invisible to the detection head.

3.1.5. Standardized Data Pre-processing Pipeline

To maintain experimental consistency across the various model scales (Nano and Small) and versions (v8 to v26), all COCO frames undergo a standardized pre-processing pipeline. First, all images are normalized to a resolution of **640 × 640 pixels**, matching the native input requirements of the YOLO family. Following resolution normalization, the artificial noise layers—Gaussian, Salt & Pepper, or Poisson—are mathematically injected at the pixel level. Finally, pixel values are scaled to a float range between 0 and 1. This normalization ensures that the detection heads receive consistent data ranges, allowing the research to focus strictly on **Architectural Robustness** rather than variance in image formatting or intensity.

3.2. Mathematical Noise Models

Three noise types were implemented.

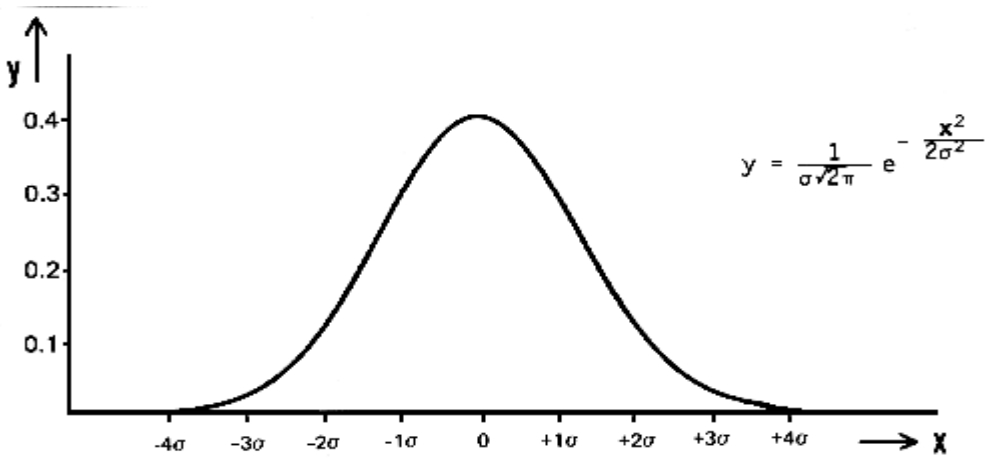
3.2.1. Gaussian Noise

Gaussian noise, also known as normal noise, follows a normal distribution with a mean (μ) of zero and a standard deviation (σ) of one. It is characterized by the bell curve shape of the distribution.

Gaussian noise is a type of noise that is generated by adding random values that are normally distributed with a mean of zero and a standard deviation (σ) to the input data. The normal distribution, also known as the Gaussian distribution, is a continuous probability distribution that is defined by its probability density function (PDF):

Gaussian Distribution formula :

$$f(x,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$



This noise follows a normal (Gaussian) distribution, adding a random value to every pixel. It is defined by its mean (μ, set to 0) and standard deviation (σ).

$$I_{noisy}(x, y) = I(x, y) + \mathcal{N}(0, \sigma^2)$$

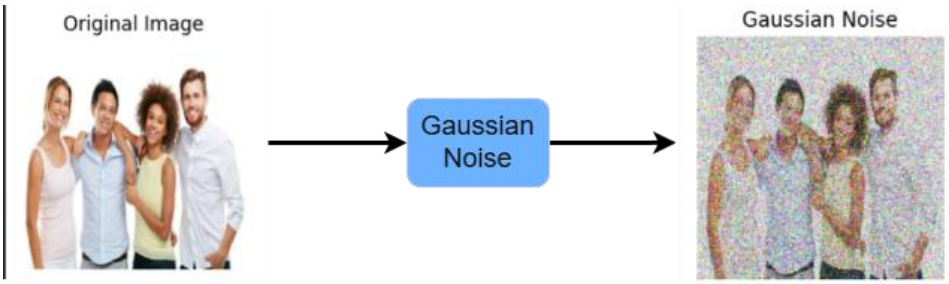


Figure 7: Gaussian Normal distribution (top), Effect of gaussian noise on pixels of image (bottom).

3.2.2. Salt & Pepper Noise

It is found only in grayscale images (black and white image). As the name suggests salt (white) in pepper (black)–white spots in the dark regions or pepper (black) in salt (white)–black spots in the white regions. Salt and pepper noise is an impulse type of noise in images. We consider salt-and-pepper noise, for which a certain amount of the pixels in the image are either black or white (black or white dots). Normally if there is black dots in the image we called it pepper noise and if there is white dots in the image we called it salt noise.

(a)

(b)

Figure 8: Effect of noise augmentation on pixels. Salt (a), pepper (b).

This impulse noise randomly replaces a fraction of pixels with pure black (0, "pepper") or pure white (255, "salt"). It is defined by the probability (p) of a pixel being corrupted.

$$I_{noisy}(x,y) = \begin{cases} 0 & with\ probability\ p/2 \\ 255 & with\ probability\ p/2 \\ I(x,y) & with\ probability\ 1 - p \end{cases}$$

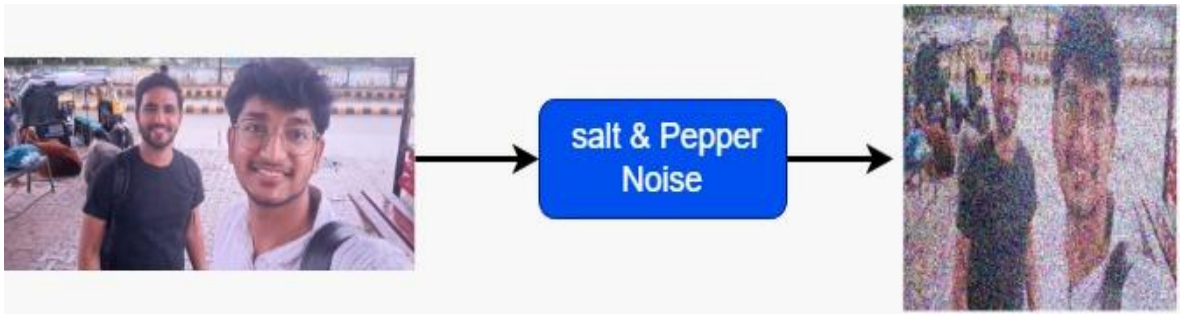


Figure 9 : Effect of Salt & Pepper noise on image.

3.2.3. Poisson Noise

Poisson noise is one type of noise that can happen when we take pictures or create digital images using sensors that detect light. It happens because light itself is made up of tiny particles called photons, and each sensor can only count a certain number of these photons before it gets full. So, if there aren't enough photons hitting the sensor, the image can end up looking noisy or grainy.

Poisson Distribution: In mathematical terms, Poisson distribution describes the probability of a given number of events happening in a fixed interval of time or space. If the average number of events (like raindrops or photons) is λ (lambda), the probability of observing k events is given by the formula :

$$\boldsymbol{P(k;\ \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}}$$

This formula helps us understand how likely it is to get a certain number of events when the average rate (λ) is known.

Also known as shot noise, this noise is signal-dependent. The noise level is proportional to the square root of the pixel intensity, making it much more prominent in bright regions. It is characteristic of photon-counting limitations in sensors.

$$I_{noisy} \sim Poisson(I(x,y)\ \times\ \lambda)$$

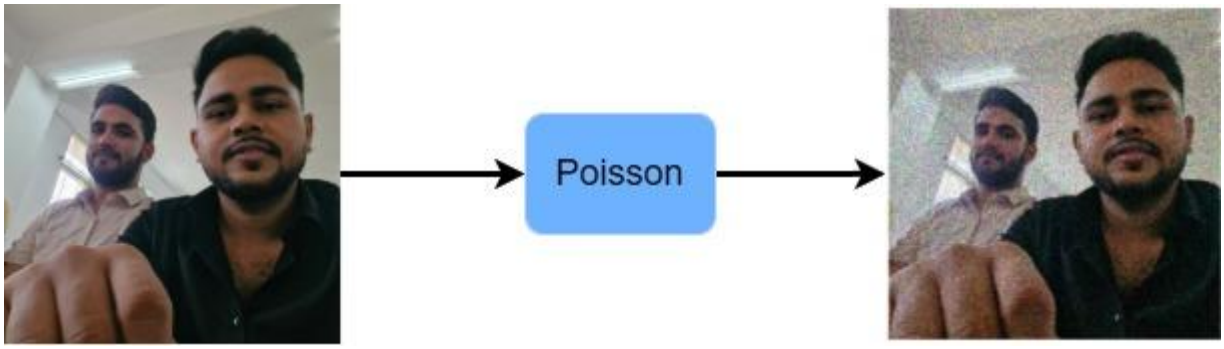


Figure 9: Effect of Poisson noise augmentation on pixels.

Here, λ is an intensity factor. In our test, a lower λ corresponds to higher noise, simulating poorer light conditions.

3.3. Evaluation Metrics

To rigorously assess the robustness of the YOLO family under varying noise conditions, a comprehensive and multi-dimensional evaluation framework is employed. This framework is designed to capture not only the **detection accuracy**, but also the **localization precision** and **operational reliability** of each model under degraded visual environments. By incorporating multiple complementary metrics, the evaluation provides a holistic understanding of model performance under environmental perturbations such as Gaussian, Salt-and-Pepper, and Poisson noise.

3.3.1. Mean Average Precision (mAP)

Mean Average Precision (mAP) is the most widely adopted metric for evaluating object detection models, as it simultaneously accounts for both **precision** (correctness of detections) and **recall** (completeness of detections). It is computed as the mean of the **Average Precision (AP)** values across all object classes, where AP represents the area under the **Precision–Recall (P–R) curve**.

The P-R curve is generated by varying the confidence threshold of detections, thereby illustrating the trade-off between precision and recall. A higher area under this curve indicates superior detection performance.

In this study, two standard variants of mAP are utilized:

- **mAP@50 (IoU = 0.5):**
  This metric evaluates detection accuracy at a fixed Intersection over Union (IoU) threshold of 0.5. A predicted bounding box is considered correct if its overlap with the ground truth exceeds 50%. This threshold is relatively lenient and primarily reflects the model's ability to detect objects, even if localization is not perfectly precise.

- **mAP@50–95 (IoU = 0.5:0.95):**

This metric provides a more comprehensive and stringent evaluation by averaging mAP over multiple IoU thresholds ranging from 0.5 to 0.95 in increments of 0.05. It emphasizes the model's capability to achieve high-precision localization, requiring predicted bounding boxes to closely align with ground truth annotations.

In the context of our work, mAP@50–95 is particularly significant, as it reflects the model's robustness to **spatial distortions introduced by noise**. For instance, under Gaussian noise, object boundaries become blurred, making precise localization challenging. A model that maintains a high mAP@50–95 under such conditions demonstrates strong resilience in accurately delineating object boundaries despite visual degradation.

3.3.2. Recall and Precision

Precision and Recall are fundamental evaluation metrics used to quantify the performance of object detection models by analyzing their prediction correctness and completeness.

- **Precision** measures the proportion of correctly predicted positive detections among all predicted detections:

$$Precision = \frac{TP}{TP + FP}$$

- **Recall** measures the proportion of correctly detected objects among all ground truth instances:

$$Recall = \frac{TP}{TP + FN}$$

where:

- TP(True Positives): Correctly detected objects
- FP(False Positives): Incorrect detections (false alarms)
- FN(False Negatives): Missed detections

In object detection tasks under noisy environments, these metrics provide critical insights into model behavior. High levels of noise may introduce **false positives** due to distorted features or increase **false negatives** when objects become indistinguishable from the background.

Precision is particularly sensitive to **spurious detections**, while recall reflects the model's ability to **retain detection capability** under degraded conditions. Therefore, analyzing both metrics together enables a comprehensive understanding of how noise impacts detection reliability.

3.3.3. Intersection over Union (IoU)

Intersection over Union (IoU) is a standard metric used to evaluate the **localization accuracy** of object detection models. It quantifies the overlap between the predicted bounding box and the ground truth

bounding box.

Mathematically, IoU is defined as:

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} = \frac{Area(B_p \cap B_{gt})}{Area(B_p \cup B_{gt})}$$

where:

- The **intersection** represents the common area between the predicted and ground truth bounding boxes.
- The **union** represents the total area covered by both bounding boxes combined.

IoU values range from 0 to 1, where:

- IoU $= 1$ indicates perfect overlap
- IoU $= 0$ indicates no overlap

In practical evaluation, a detection is considered correct if its IoU exceeds a predefined threshold (commonly 0.5).

In this study, IoU plays a critical role in assessing the impact of noise on **spatial localization accuracy**. For instance, Gaussian noise can blur object boundaries, while Salt-and-Pepper noise introduces abrupt pixel variations, both of which can degrade bounding box alignment. Consequently, maintaining high IoU under noisy conditions indicates strong localization robustness.

3.3.4. F1-Score

The F1-Score is a widely used evaluation metric that provides a balanced measure of a model's **precision** and **recall**, particularly in scenarios where both false positives and false negatives are critical. It is defined as the harmonic mean of precision and recall, ensuring that a model achieves high performance only when both components are sufficiently strong.

Mathematically, the F1-Score is expressed as:

$$F1 = 2 \times \frac{Precision\ \times Recall}{Precision\ +\ Recall}$$

where precision represents the proportion of correctly predicted positive detections among all predicted detections, and recall represents the proportion of correctly detected objects among all ground truth instances.

In the context of object detection under noisy conditions, the F1-Score plays a crucial role in evaluating the trade-off between missed detections and false alarms. For example, under high-intensity noise, models

may either fail to detect objects (reducing recall) or produce spurious detections (reducing precision). The F1-Score effectively captures this balance in a single scalar value.

A higher F1-Score indicates that the model maintains both high detection accuracy and reliability, even when image quality is degraded. Therefore, it serves as an essential complement to mAP by highlighting the model's overall detection consistency rather than focusing solely on localization precision.

3.3.5. Latency and Efficiency Metrics

In addition to accuracy and robustness, the **computational efficiency** of object detection models is a critical factor, especially for real-time and resource-constrained applications. To evaluate this aspect, latency and confidence-based metrics are incorporated into the analysis.

- **Inference Latency (ms):**
  Latency measures the average time required by a model to process a single image and produce predictions. It is computed in milliseconds and serves as an indicator of the model's suitability for real-time deployment. Lower latency values correspond to faster models, which are preferable in applications such as surveillance, autonomous systems, and edge computing.
- **Mean Detection Confidence:**
  This metric represents the average confidence score of detected objects across all predictions. It provides insight into the model's certainty in its detections. Under noisy conditions, a decline in confidence may indicate increased ambiguity in feature extraction and object representation.

In this study, latency is measured under moderate noise conditions to simulate realistic operational environments, ensuring that the evaluation reflects practical deployment scenarios. The combination of latency and confidence enables a comprehensive assessment of the **efficiency–reliability trade-off**.

Furthermore, an **Efficiency–Robustness Frontier** is constructed by plotting inference latency against performance (e.g., mAP at a moderate noise level). This visualization facilitates the identification of models that achieve an optimal balance between speed and robustness, thereby aiding in the selection of the most suitable model for specific real-world applications.

3.3.6. Robustness Retention Index (RRI)

To further quantify the stability of model performance under noise, the **Robustness Retention Index (RRI)** is introduced. Unlike the Robustness Index, which measures performance degradation, RRI evaluates the proportion of performance retained relative to the clean (noise-free) condition.

The RRI is defined as:

$$RRI = \frac{mAP_{Noisy}}{mAP_{Clean}} \times 100$$

where:

- mAP_{clean} is the model performance at zero noise level
- mAP_{noisy} is the performance at a specified noise level (typically the highest level)

The RRI is expressed as a percentage, where:

- Higher values indicate better robustness (greater retention of performance)
- Lower values indicate significant degradation under noise

This metric provides an intuitive interpretation of model resilience. For example, an RRI of 80% implies that the model retains 80% of its original performance under adverse conditions.

The RRI is particularly useful for comparative evaluation across models, as it highlights those that maintain **consistent detection capability despite environmental distortions**. When analyzed alongside the Robustness Index, it offers a dual perspective—capturing both performance loss and performance retention—thereby enabling a more comprehensive robustness assessment.

3.4. Implementation and Experimental Setup

All experiments were conducted in a controlled environment using Python 3.10 integrated with the Ultralytics YOLO framework. The implementation leveraged standard scientific computing and image processing libraries, including OpenCV for synthetic noise injection, NumPy for efficient numerical and matrix operations, and Matplotlib for performance visualization and degradation analysis.

A comprehensive multi-model evaluation framework was designed to assess robustness across different architectural scales. Specifically, the Nano (n) and Small (s) variants of YOLOv8, YOLOv9, YOLOv10, YOLOv11, and YOLOv26 were selected to represent lightweight and moderately complex detection models. All models were initialized with pre-trained weights of the COCO-trained dataset to emulate realistic *out-of-the-box deployment scenarios*, thereby avoiding bias introduced by task-specific fine-tuning.

For inference the standardized detection parameters were employed across all models to ensure fair comparison. The confidence threshold was set to 0.5 which helps filtering out low-confidence predictions and improving detection reliability. Additionally, the Intersection over Union (IoU) threshold was fixed at 0.3 which allows flexibility in bounding box matching under noisy conditions where precise localization may degrade.

To systematically evaluate robustness that is a stochastic noise injection pipeline was implemented. Three distinct noise models—Gaussian noise, Salt-and-Pepper noise, and Poisson noise—were applied to the input images. Each noise type was introduced progressively across six discrete noise levels from level 0

to level 5 representing a controlled transition from noise-free (pristine) conditions to highly degraded (catastrophic) environments. This iterative augmentation strategy enabled a fine-grained analysis of model performance decay as a function of increasing noise intensity.

The experiment is designed so that results can be repeated easily, all models are tested fairly, and their performance is checked under different levels of increasing noise. The different noise intensity used in the workflow of the project are listed in the table below for all the three noises.

Level	**Gaussian (σ)**	**Salt & Pepper (ρ)**	**Poisson (λ)**
0 (Clean)	0.0	0.0	0.0
1	0.1	0.01	0.01
2	0.2	0.05	0.05
3	0.5	0.1	0.1
4	1	0.3	1
5 (Extreme)	2	0.5	10

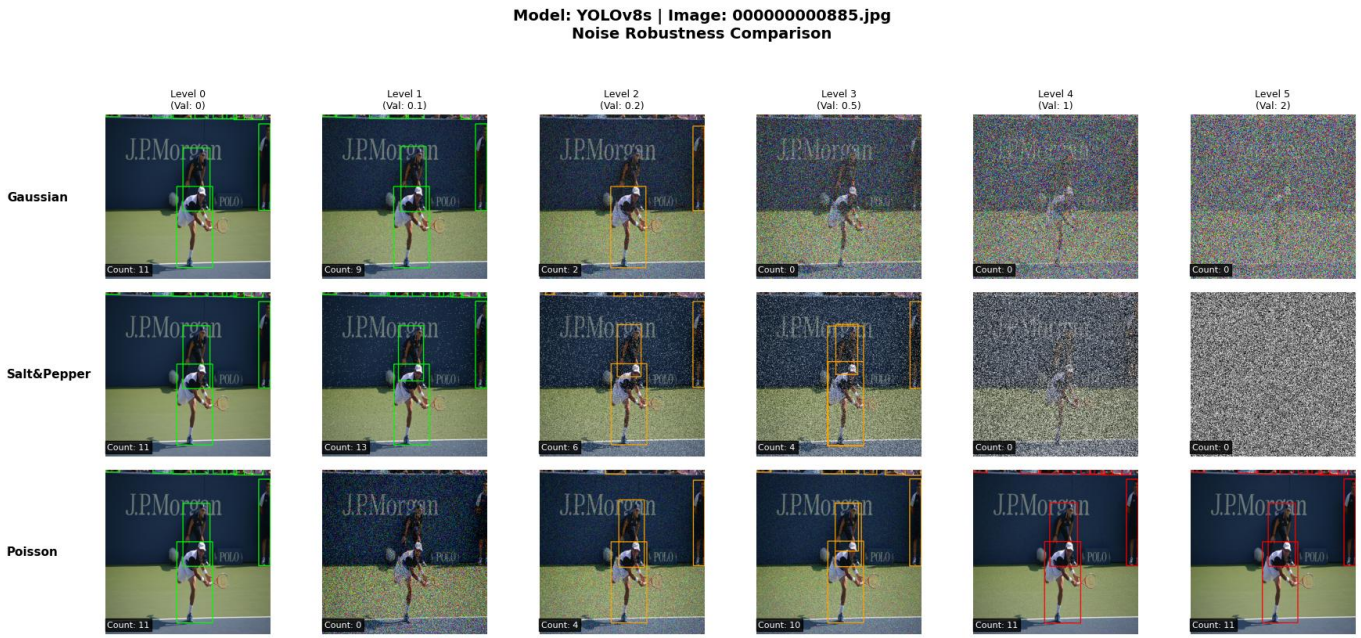
Table 3: Noise levels of different noises.

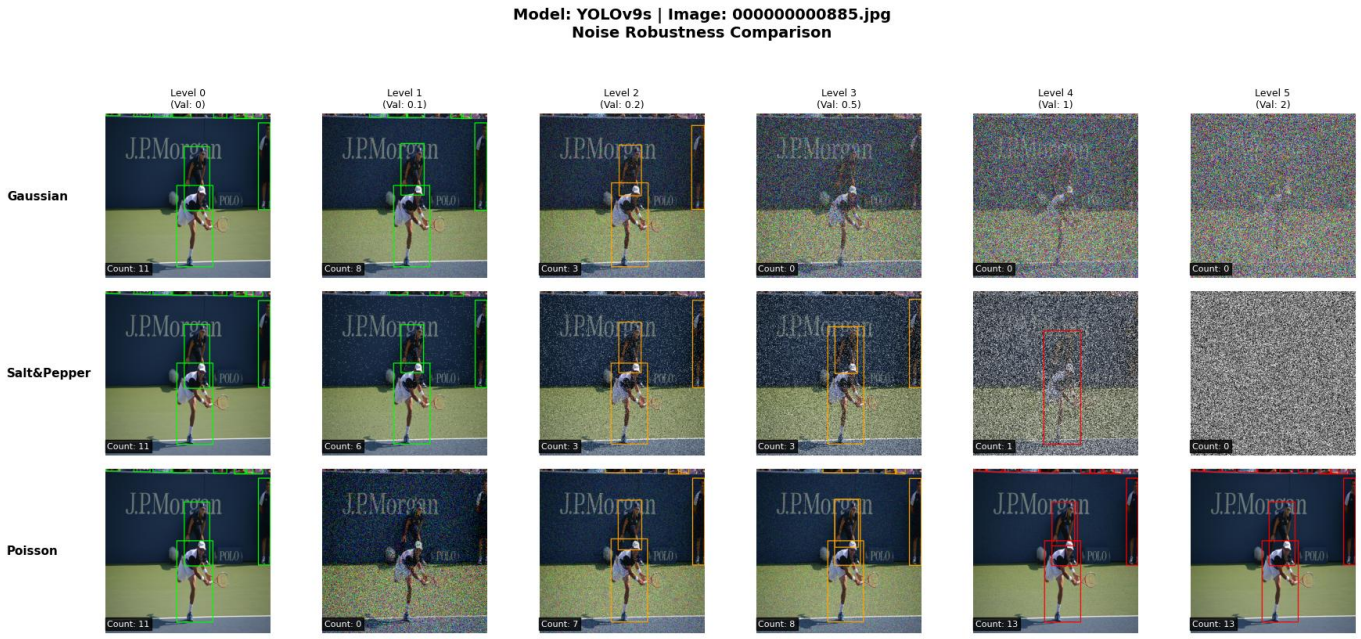
Note on Poisson Noise: In the Poisson model, λ represents the photon count. Therefore, a **lower λ** (e.g., 0.01) represents a **higher** noise intensity (extreme darkness).

4.0 Results and Analysis

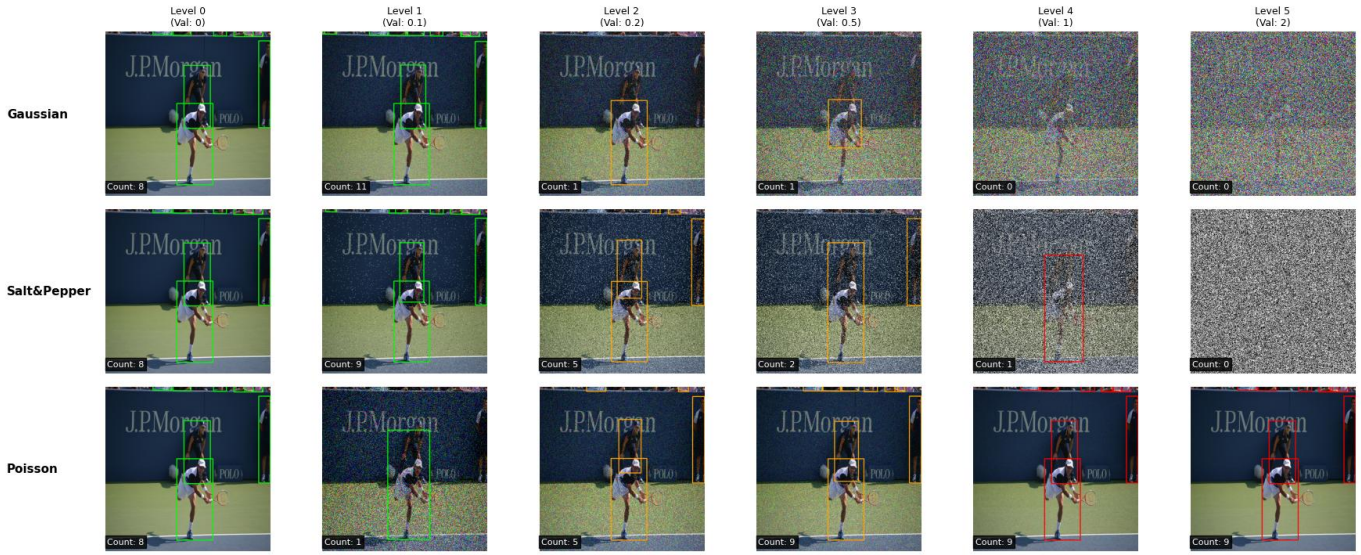
4.1 Robustness Analysis under Noise

The performance of each model was evaluated using the percentage drop in mAP and a normalized robustness score across Gaussian, Salt & Pepper, and Poisson noise. The robustness of various YOLO architectures was systematically evaluated under three distinct types of image degradation: Gaussian noise, Salt & Pepper noise, and Poisson noise. The evaluation was conducted using mAP@0.5 across multiple noise intensities, enabling a comprehensive understanding of how detection performance degrades under realistic perturbations. The effect of noise on the performance of different models is as follows:

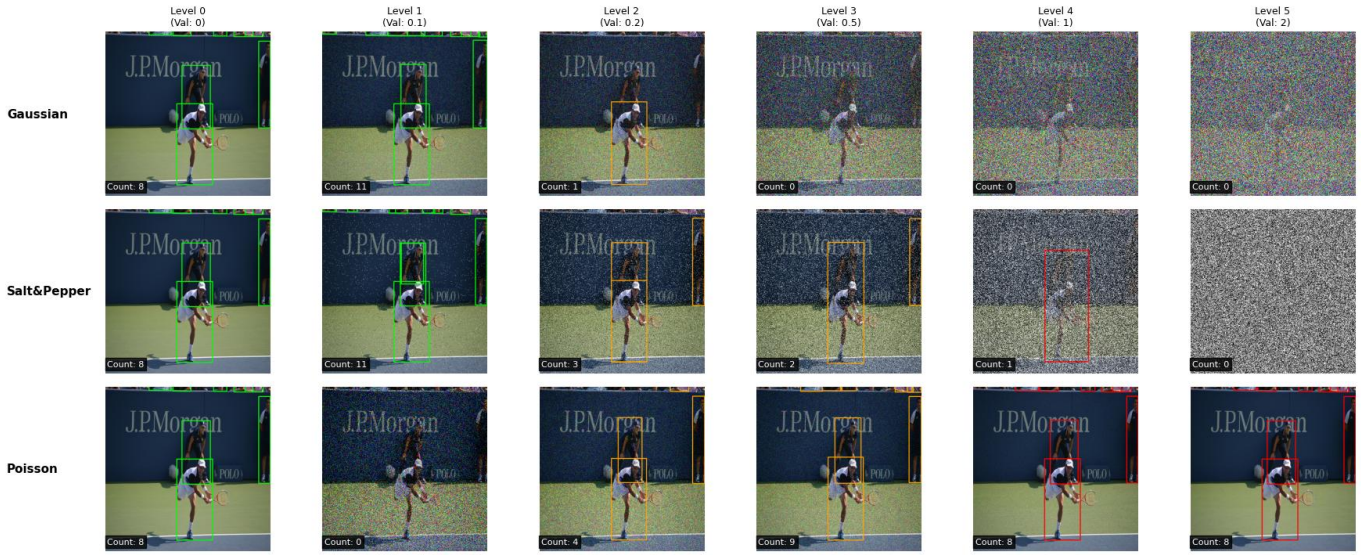




Model: YOLOv10s | Image: 000000000885.jpg
Noise Robustness Comparison



Model: YOLO11s | Image: 000000000885.jpg
Noise Robustness Comparison



Model: YOLO26s | Image: 000000000885.jpg
Noise Robustness Comparison

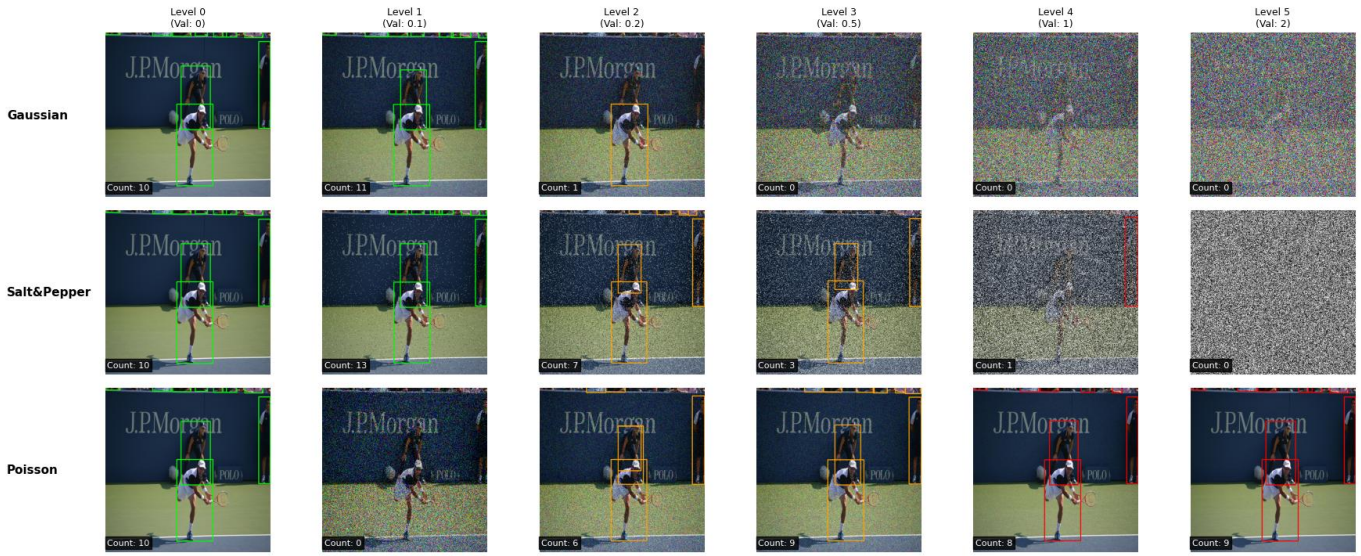


Figure 10: The effect of noise over the performance of object detection for YOLO models (YOLOv8n, YOLOv9n, YOLO10n, YOLOv11n, YOLO26n) top to bottom respectively.

MAP vs noise intensity

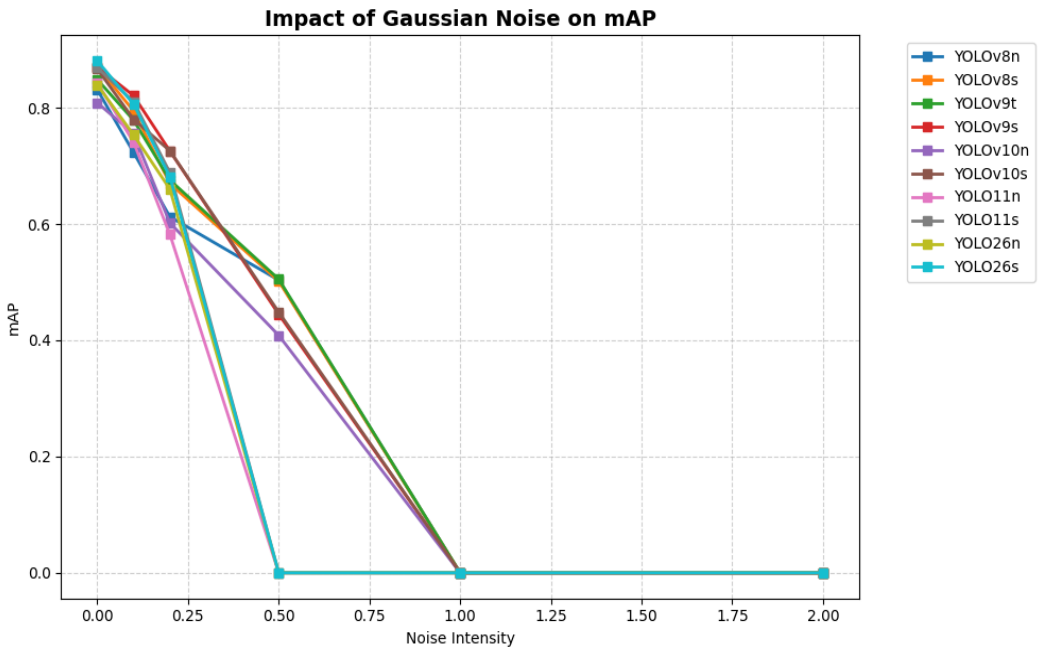


Figure 11: Impact of Gaussian Noise on mAP for all the models.

The results indicate that **Gaussian noise has the most severe impact on object detection performance**. As observed in the Gaussian mAP curves, all models exhibit a steep decline in performance as noise intensity increases. At low noise levels ($\sigma \leq 0.1$), models maintain relatively high detection accuracy, indicating resilience to minor perturbations. However, as the noise intensity increases to moderate levels ($\sigma \approx 0.2$–0.5), a sharp degradation is observed across all architectures. This degradation becomes catastrophic at higher levels ($\sigma \geq 1$), where mAP approaches zero. This behavior can be attributed to the nature of Gaussian noise, which uniformly distorts pixel intensities across the entire image, thereby disrupting edge structures and spatial features essential for convolutional feature extraction.

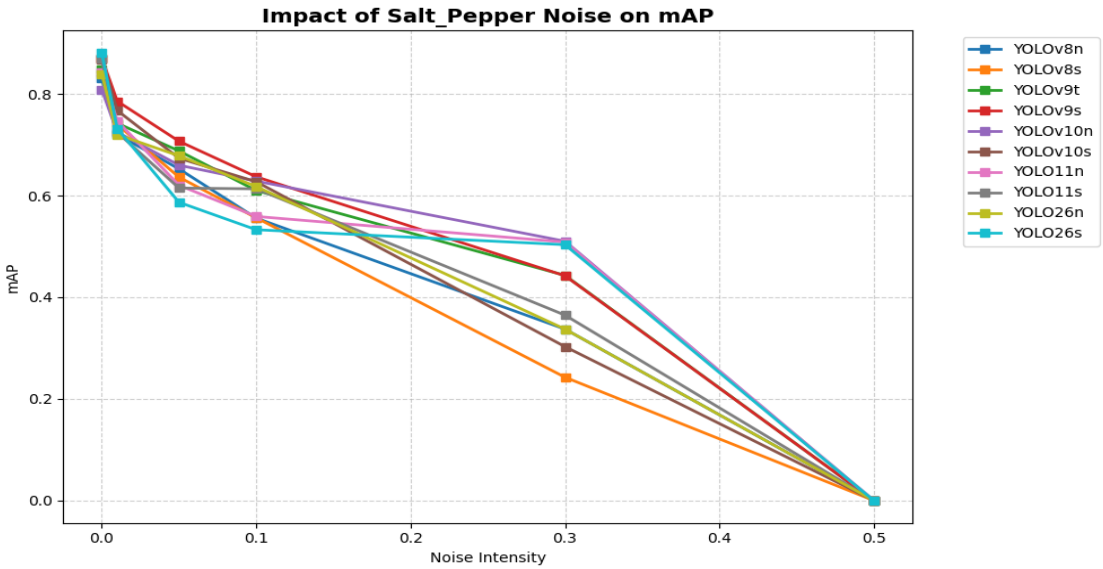


Figure 12: Impact of Salt & Pepper Noise on mAP for all the models.

In contrast, **Salt & Pepper noise exhibits a comparatively gradual degradation pattern**. Since this noise type affects only a subset of pixels by introducing random black and white impulses, a significant portion of the image structure remains intact. As a result, models retain moderate detection capability up to intermediate noise levels (0.1–0.3). However, beyond this threshold, the density of corrupted pixels increases, leading to a progressive loss of structural information and eventual collapse in performance at extreme noise levels (0.5). This gradual decay highlights that YOLO models are relatively more tolerant to sparse noise compared to dense noise distributions.

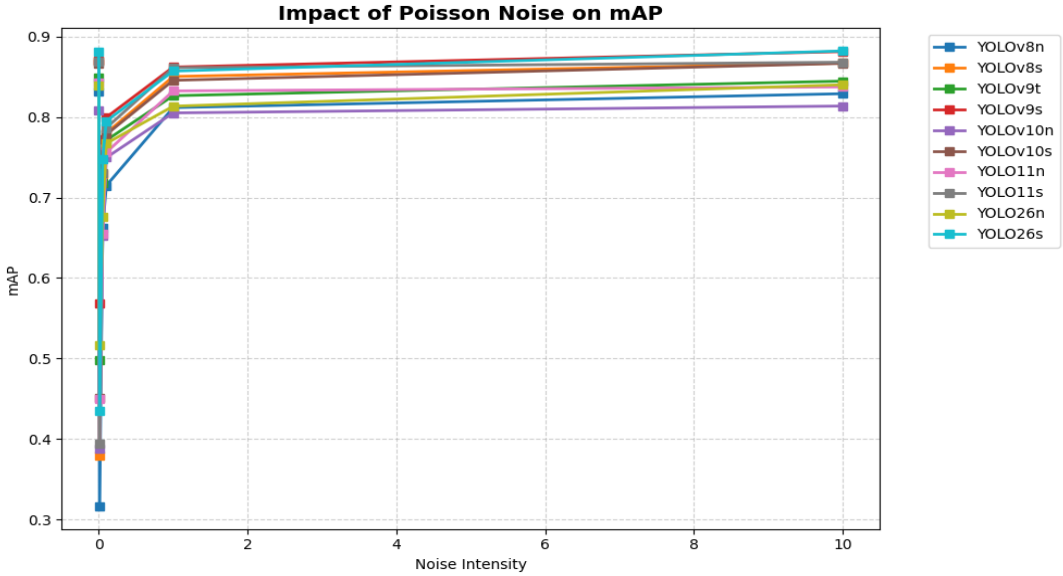


Figure 13: Impact of Gaussian Noise on mAP for all the models.

Interestingly, **Poisson noise demonstrates minimal adverse impact on detection performance**, and in some cases, even results in slight improvements at low noise levels. This phenomenon can be attributed to the signal-dependent nature of Poisson noise, which preserves relative intensity variations and does not significantly distort structural patterns. Consequently, feature extraction remains largely unaffected, allowing models to maintain stable mAP values across all noise levels. This observation suggests that YOLO architectures are inherently robust to certain stochastic noise distributions that preserve underlying image characteristics.

Among all evaluated models, **YOLOv9s consistently exhibits the highest robustness across Gaussian and Salt & Pepper noise conditions**, maintaining superior mAP values even at moderate noise levels. This can be attributed to its deeper architecture and enhanced feature representation capabilities, which enable better generalization under degraded inputs.

4.2 F1-Score Analysis

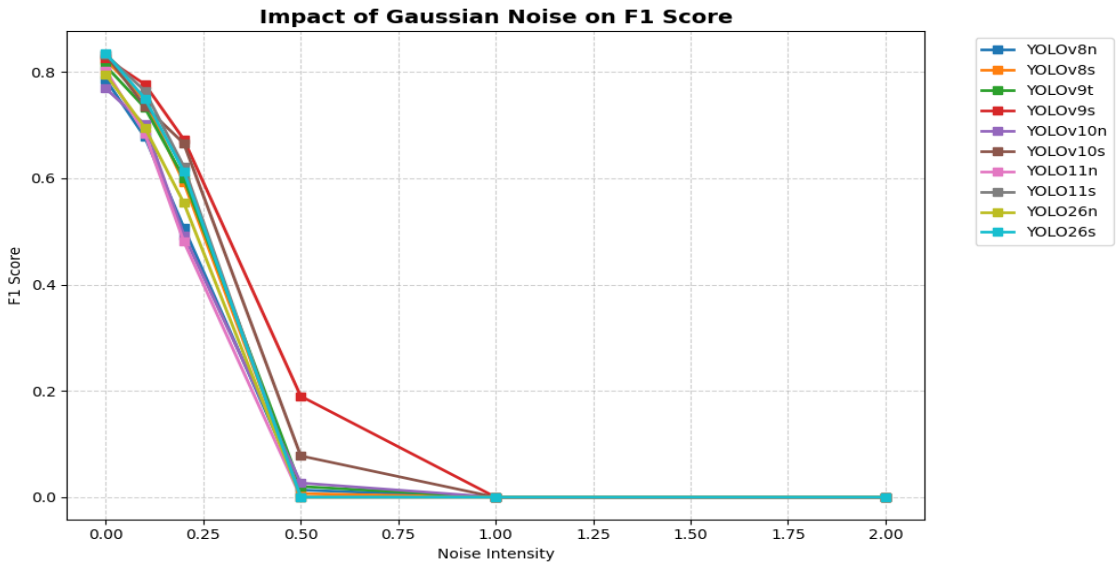
To further analyze detection quality, the F1-score was evaluated as a combined measure of precision and recall. The F1-score trends closely align with the mAP observations, providing deeper insight into the behaviour of the models under noisy conditions.

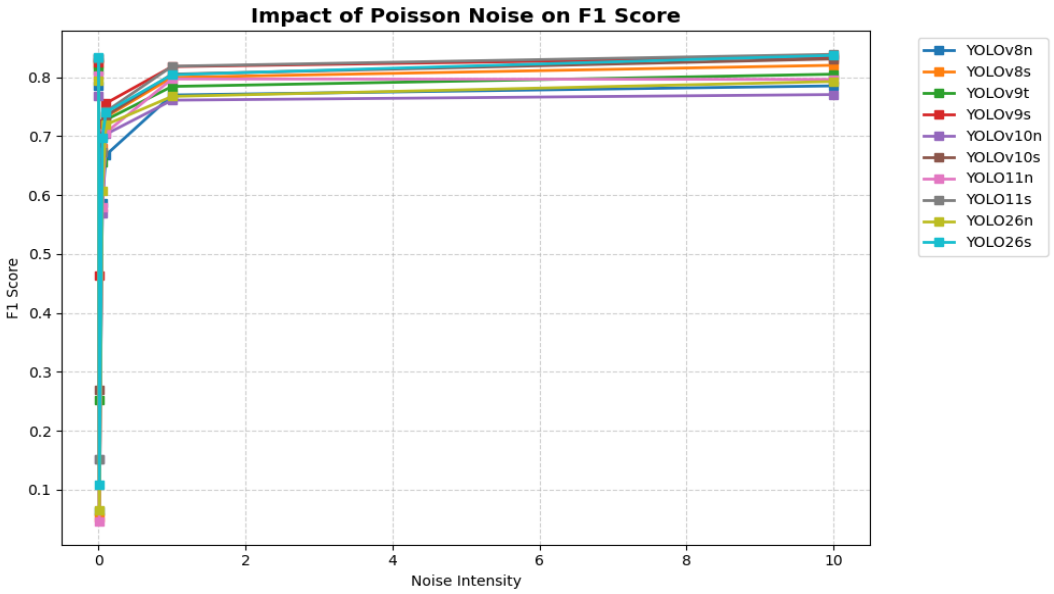
Under Gaussian noise, the F1-score exhibits a rapid decline beyond moderate noise levels, indicating a simultaneous degradation in both precision and recall. This suggests that Gaussian noise not only affects object localization but also significantly impacts classification confidence. Notably, at higher noise levels, the F1-score approaches zero, indicating complete failure in detection.

For Salt & Pepper noise, the F1-score decreases more gradually, mirroring the behaviour observed in mAP. The partial preservation of image structure allows models to maintain both detection and classification performance up to moderate noise intensities. However, as noise density increases, both precision and recall deteriorate, leading to a sharp drop in F1-score.

In the case of Poisson noise, the F1-score remains relatively stable across all models and noise levels. This further confirms that Poisson noise does not significantly disrupt feature extraction or detection confidence.

Overall, the F1-score analysis reinforces the conclusion that **Gaussian noise is the most destructive**, while **Poisson noise has negligible impact**, thereby validating the robustness trends observed in mAP.





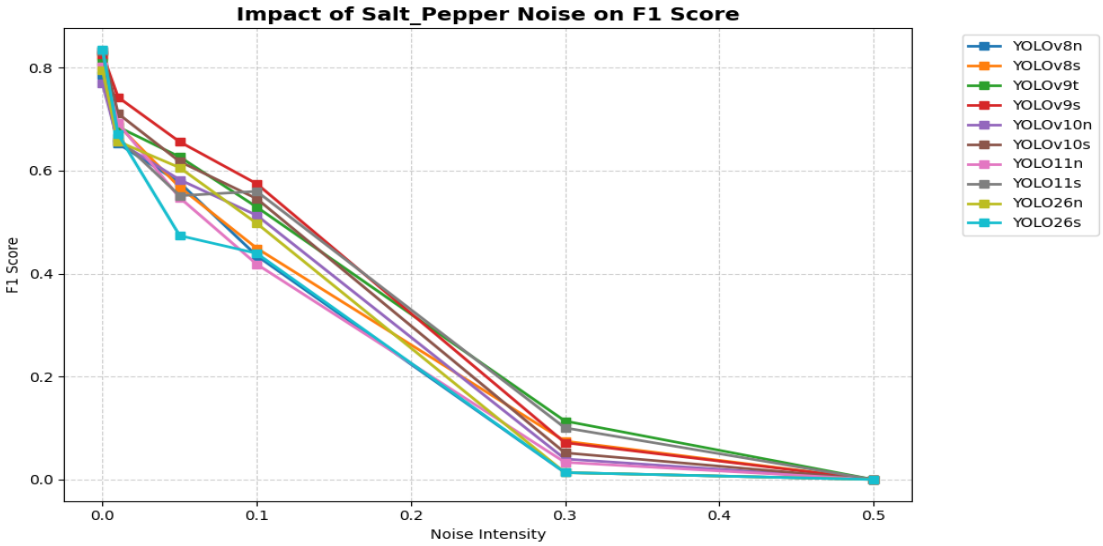


Figure 14: The degradation of models mAP@0.5 with varying noise intensities over different noise types. Gaussian (Top), Salt & pepper (middle) and Poisson (Bottom).

4.3 Robustness Retention Index (RRI) Analysis

To quantitatively evaluate the robustness of the models relative to their baseline performance, the **RRI** was computed. The RRI represents the ratio of the model's performance under noisy conditions to its performance on clean images, thereby providing a normalized measure of robustness.

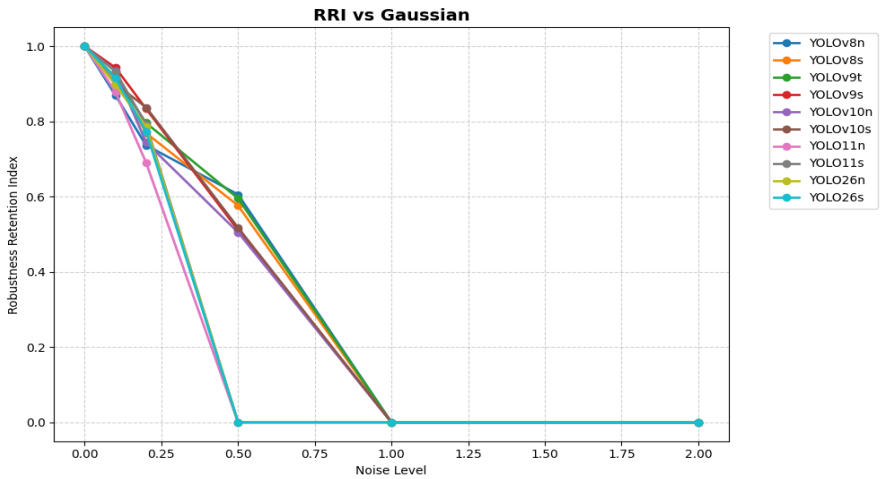
Figure 14 illustrate the degradation behaviour of different YOLO models across varying noise intensities and types. Under **Gaussian noise**, all models exhibit a sharp decline in RRI as noise intensity increases. At moderate noise levels, the RRI drops significantly, indicating a substantial loss of detection capability relative to clean conditions. This trend confirms that Gaussian noise severely affects feature extraction, leading to rapid degradation in model performance.`

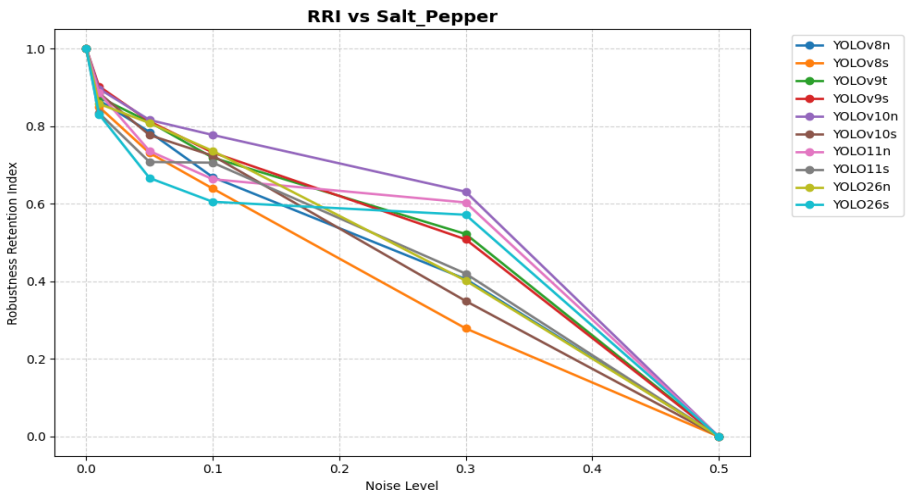
In contrast, under **Salt & Pepper noise**, the RRI decreases more gradually. The models retain a higher fraction of their original performance at intermediate noise levels, reflecting their ability to handle sparse pixel corruption. However, at higher noise densities, the RRI approaches zero, indicating eventual failure in detection.

For **Poisson noise**, the RRI remains consistently high across all models and noise levels, often close to unity. This suggests that the models retain most of their original performance, further validating the minimal impact of Poisson noise on detection accuracy.

Among all models, **YOLOv9s and YOLOv10n demonstrate the highest robustness retention**, maintaining relatively higher RRI values across Gaussian and Salt & Pepper noise conditions. In contrast, models such as YOLO11s and YOLO26s show a faster decline in RRI, indicating weaker resilience to noise.

Overall, the RRI analysis provides a normalized perspective of robustness, complementing the mAP and F1-score evaluations. It reinforces the conclusion that Gaussian noise is the most disruptive, while Poisson noise has negligible impact, and highlights the superior robustness of specific architectures such as YOLOv9s.





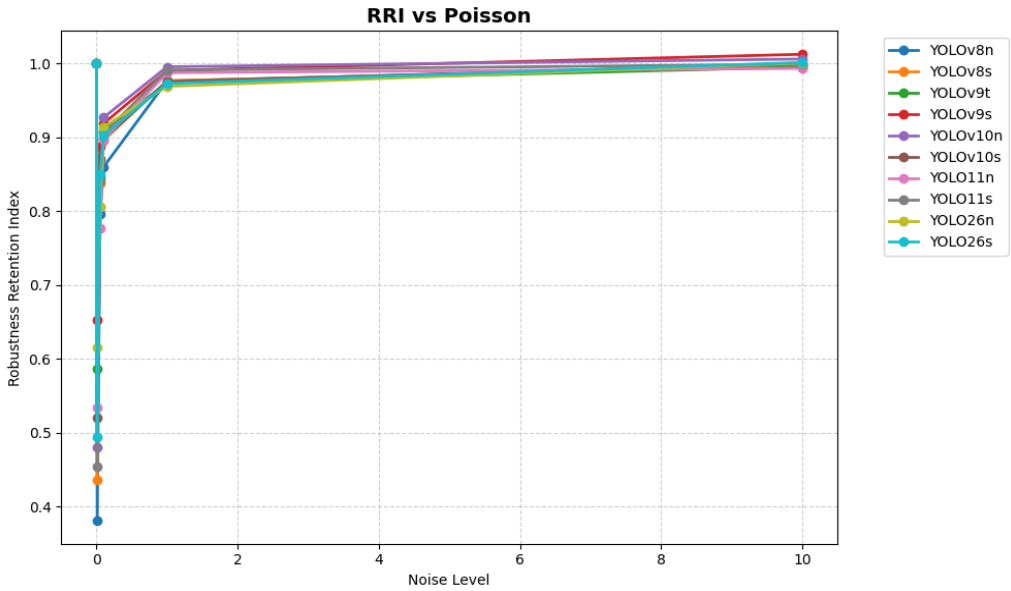


Figure 15: Effect of noise levels on the Robustness Retention Index. Gaussian noise (top), Salt & Pepper noise (middle) and Poisson noise (bottom).

4.4 Quantitative Performance

A comprehensive quantitative comparison of all models is presented in Table 5, which summarizes the average performance degradation, mean precision, mean F1-score, and robustness retention index (RRI).

From Table 4, **YOLOv9s achieves the highest detection accuracy and robustness**, with the lowest average drop (35.28%) and highest mean F1-score (0.5419). This indicates its superior ability to maintain performance under varying noise conditions. Additionally, it achieves the highest RRI (≈0.706) showing the robustness of YOLOv9s over augmented noise.

YOLOv10n emerges as the most balanced model, achieving a strong RRI (≈0.696) while maintaining significantly lower latency compared to larger models. Its lightweight architecture enables efficient computation without substantial compromise in robustness, making it highly suitable for real-time applications.

Models such as **YOLOv9t and YOLOv10s** also demonstrate competitive performance, with relatively high precision and F1-scores. However, their higher computational cost reduces their practicality in latency-sensitive environments.

On the other hand, YOLO11s, YOLO26s, and YOLO11n exhibit higher performance degradation (>42%), indicating weaker robustness under noisy conditions. This suggests that their architectures are less effective in preserving feature representations when subjected to image degradation.

Table 4.4.1 YOLO Model Performance under Noise

The overall detection performance of the evaluated YOLO models under noisy conditions is summarized in Table 4. The table presents the average percentage drop in mAP along with mean precision and F1-score across all noise types and levels. These metrics provide a consolidated view of model robustness and detection accuracy.

Model	Avg Drop (%)	Mean Precision	Mean F1
YOLOv8n	40.33098	0.687712	0.450274
YOLOv8s	40.67097	0.676711	0.482094
YOLOv9t	36.35169	0.725032	0.498099
YOLOv9s	35.28268	0.704139	0.541931
YOLOv10n	36.52721	0.721698	0.4596
YOLOv10s	38.29623	0.696579	0.513431
YOLO11n	42.35075	0.66529	0.45501
YOLO11s	42.78138	0.643837	0.499008
YOLO26n	41.37061	0.663671	0.464377
YOLO26s	42.77135	0.654838	0.480597

Table 4: Summary of YOLO Model Performance under Noise Conditions

From Table 4.1, it can be observed that YOLOv9s achieves the lowest average performance drop (35.28%) and the highest F1-score (0.5419), indicating superior robustness and detection capability under noisy conditions. Models such as YOLOv9t and YOLOv10n also demonstrate competitive performance with relatively lower degradation.

In contrast, models like YOLO11s, YOLO26s, and YOLO11n exhibit higher performance degradation (>42%), indicating reduced robustness. Additionally, while lightweight models such as YOLOv8n provide faster inference, they show comparatively lower F1-scores, highlighting the trade-off between efficiency and accuracy.

4.5 Efficiency and Latency Analysis

The computational efficiency of each model was evaluated using inference latency, as illustrated in the latency comparison plot.

The results show that **YOLOv10n (111.56 ms), YOLO11n (114.54 ms), and YOLOv8n (119.03 ms)** are the fastest models, making them suitable for real-time and embedded applications. These models

achieve low latency due to their lightweight architectures and reduced parameter complexity.

In contrast, **YOLOv9t (353.00 ms), YOLO26s (302.20 ms), and YOLOv9s (298.88 ms)** exhibit significantly higher latency. While these models offer improved robustness and accuracy, their computational overhead limits their applicability in real-time systems.

This analysis highlights a fundamental trade-off: **increased model complexity improves robustness but incurs higher computational cost**, whereas lightweight models achieve faster inference at the expense of robustness.

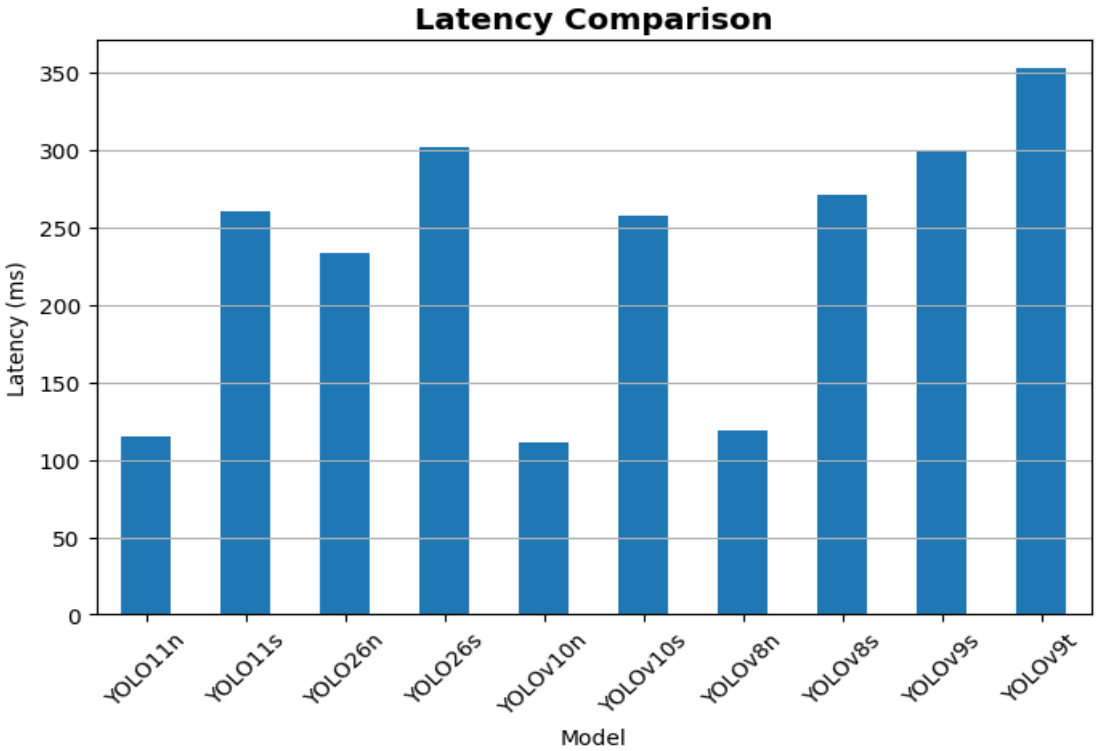


Figure 16: Latency of YOLO models.

4.6 Efficiency–Robustness Trade-off Analysis

The trade-off between robustness and efficiency is visualized using the efficiency–robustness scatter plot.

The results indicate that **YOLOv10n occupies the optimal region**, achieving a strong balance between low latency and high robustness. This makes it the most practical model for real-world applications requiring both speed and reliability.

YOLOv9s is positioned in the high-robustness but low-efficiency region, offering superior detection performance but at the cost of increased latency. Conversely, models such as YOLOv8n and YOLO11n lie in the low-latency but moderate-robustness region, making them suitable for time-critical applications.

Models like YOLO26s and YOLOv9t fall in less favourable regions due to either excessive latency or suboptimal robustness, indicating inefficiencies in their design for noisy environments.

This analysis confirms that **no single model dominates across all performance dimensions**, and the optimal choice depends on application-specific constraints.

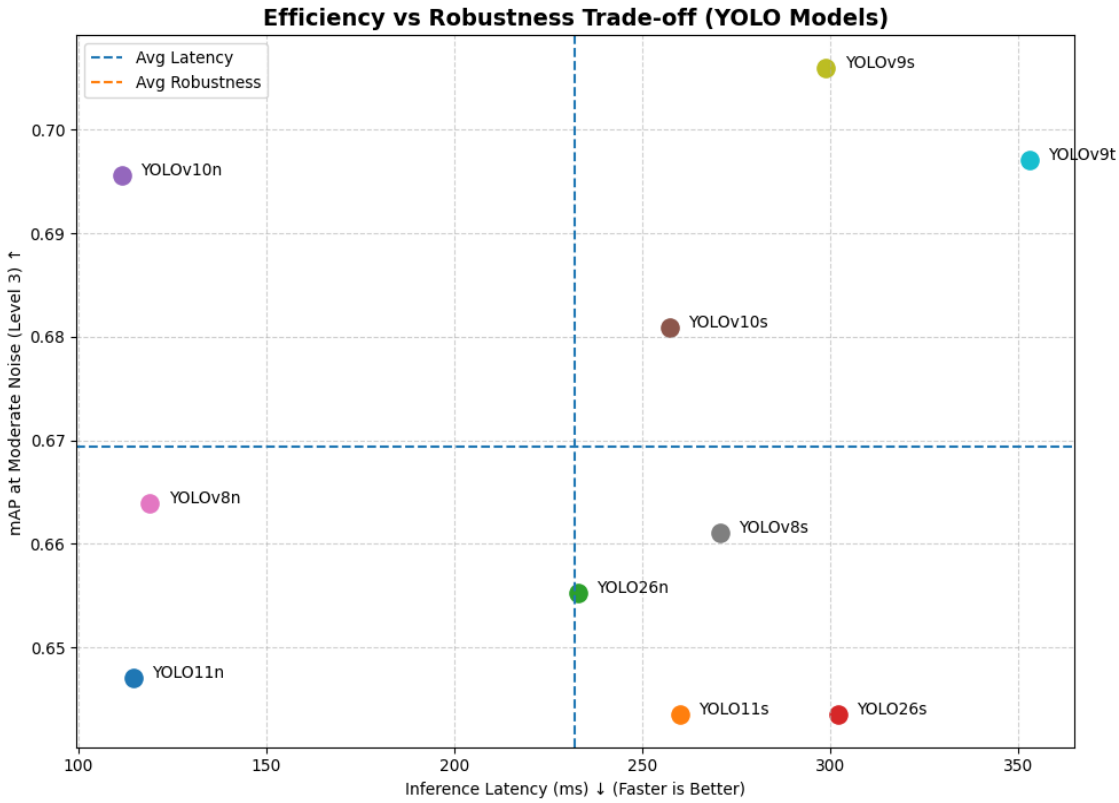


Figure 17: Robustness trade-offs of models with efficiency, YOLOv9s performing best on robustness while YOLO26s has degraded robustness with most efficiency.

4.4 Comparative Performance of YOLO Models

The overall performance of the evaluated YOLO models is summarized in Table 5 which combines robustness, accuracy, and efficiency metrics into a unified comparison. The table includes average performance degradation, mean F1-score, robustness retention index (RRI), and inference latency. A composite final score is computed to provide an integrated evaluation of each model by balancing robustness (RRI), detection accuracy (F1-score), and computational efficiency (latency).

Model	**Avg Drop (%)**	**Mean Precision**	**Mean F1**	**Avg RRI**	**Latency (ms)**	**Latency Score**	**Final Score**
YOLOv8n	40.33099	0.687712	0.450274	0.663908	119.0302	0.662811	0.578235
YOLOv8s	40.67097	0.676711	0.482094	0.661075	270.6198	0.233388	0.503945
YOLOv9t	36.35169	0.725032	0.498099	0.697069	353.0077	0	0.478067
YOLOv9s	35.28268	0.704139	0.541931	0.705978	298.8879	0.15331	0.529826
YOLOv10n	36.52721	0.721698	0.4596	0.695607	111.5679	0.68395	0.598873
YOLOv10s	38.29623	0.696579	0.513431	0.680865	257.402	0.270832	0.531885

YOLO11n	42.35075	0.66529	0.45501	0.647077	114.5437	0.675521	0.575939
YOLO11s	42.78139	0.643837	0.499008	0.643488	259.9737	0.263547	0.509708
YOLO26n	41.37061	0.663671	0.464377	0.655245	232.9925	0.339979	0.515845
YOLO26s	42.77135	0.654838	0.480597	0.643572	302.2003	0.143927	0.478453

Table 5: Comparative Performance of YOLO Models under Noise and Efficiency Constraints

From Table 5 it can be observed that YOLOv10n achieves the highest overall performance with the highest final score of 0.5989. This is primarily due to its strong balance between robustness (RRI ≈ 0.696) and low latency (≈111 ms) making it the most suitable model for real-time applications.

YOLOv9s demonstrates the highest robustness and detection accuracy achieving the lowest performance degradation (35.28%) and highest F1-score (0.5419). However its high latency (≈299 ms) limits its applicability in time-sensitive scenarios.

Models such as YOLOv8n and YOLO11n exhibit low latency which makes them suitable for real-time deployment but their relatively higher performance degradation indicates reduced robustness under noisy conditions. In contrast larger models such as YOLOv9t and YOLO26s suffer from high latency without proportional gains in robustness which indicates inefficiencies in handling degraded inputs.

Overall the results highlight a clear trade-off between robustness and efficiency confirming that the model YOLOv10n provides the best balance where as YOLOv9s is the most robust model and YOLOv8n YOLO11n are best suited for real-time applications.

4.4.1 Performance Decay Analysis

The performance decay trends reveal important insights into model behaviour under increasing noise intensity.

A notable observation is the presence of a **threshold effect in Gaussian noise** where performance drops sharply beyond $\sigma \approx 0.3$–0.5. This indicates a critical point beyond which feature extraction fails. In contrast Salt & Pepper noise results in a smoother degradation curve while Poisson noise maintains stable performance.

These findings emphasize the importance of evaluating models across a range of noise levels as extreme conditions alone do not fully capture model robustness.

Model	Strength	Weaknes s	Best Metric	Why it Performs This Way	Suitable Use Case	Overall Verdict
YOLOv10 n	Best balance (Score: 0.599), low latency	Slightly lower F1 than v9s	Efficiency + Robustness	Lightweight architecture + optimized feature extraction	Real-time + moderate noise	Best Overall Model
YOLOv9s	Highest robustness (RRI ≈ 0.706), highest F1	High latency	Robustness + Accuracy	Deeper architecture captures complex features	Noisy environments , high accuracy tasks	Most Robust Model
YOLOv9t	High precision, good RRI	Very high latency	Precision	Transformer-based features improve detection confidence	Offline / batch processing	Good but inefficient
YOLOv10s	Good F1 (0.513), stable performanc e	Medium latency	Balanced accuracy	Improved feature fusion vs v10n	Balanced systems	Reliable alternative
YOLOv8n	Fast, good latency	Lower robustness	Speed	Lightweight design limits feature depth	Edge devices, real-time	Efficient but less robust
YOLO11n	Very fast, stable	Higher drop %, lower robustness	Latency	Compact architecture prioritizes speed	Embedded systems	Fast but weaker
YOLOv8s	Moderate F1, stable	High latency vs gain	Accuracy	Larger than v8n but not significantly better	General use	Not optimal
YOLO26n	Moderate latency, stable	Lower robustness	Balanced	Architecture not optimized for noise	Standard detection	Average performer
YOLO11s	Decent F1	High drop %, slower	Moderate accuracy	Larger but not robust to noise	Non-critical tasks	Suboptimal
YOLO26s	High confidence	Worst efficiency + robustness	Confidenc e	Over-parameterized , sensitive to noise	Rare use	Least recommende d

Table 6: Comprehensive Model Comparison and Interpretation

5.0 Applications of This Project

The proposed study on the robustness and efficiency of multiple YOLO-based object detection models (YOLOv8–YOLO26 variants) under diverse noise conditions has significant implications for real-world computer vision systems where input data is frequently degraded due to sensor limitations environmental interference or transmission artifacts. By systematically analyzing the impact of Gaussian, Salt-and-Pepper, and Poisson noise the framework provides critical insights into model reliability which enables informed selection and deployment of detection architectures in noise-prone environments.

In **autonomous driving and intelligent transportation systems** the findings can be directly applied to enhance the reliability of pedestrian, vehicle, and obstacle detection under adverse conditions such as fog, rain, dust and low illumination. Improved robustness to noise contributes to safer decision-making in Advanced Driver Assistance Systems (ADAS) and fully autonomous vehicles thereby reducing the risk of accidents and improving overall traffic management efficiency.

In **security and defence applications** robust object detection is essential for surveillance and reconnaissance tasks involving UAVs, drones and night-vision systems. These systems often operate under low visibility and noisy imaging conditions including thermal and infrared domains. The insights from this study support the development of resilient detection pipelines capable of maintaining accuracy in mission-critical scenarios such as border monitoring, threat detection and disaster response.

In the **medical and healthcare domain** the proposed framework is highly relevant for improving the reliability of automated diagnostic systems that rely on inherently noisy imaging modalities such as X-rays, MRI, CT scans, and ultrasound. Enhancing detection robustness can aid in accurate anomaly detection, tumor localization, and clinical decision support. It can improve the performance of robotic-assisted surgical systems and telemedicine platforms where image quality may be compromised due to compression or transmission noise.

In **industrial automation and smart manufacturing** the findings can be utilized for defect detection, visual inspection and object tracking in production environments characterized by variable lighting conditions, mechanical vibrations, and sensor noise. Robust detection models enable higher precision in quality control processes and reduce false detections thereby improving operational efficiency and reducing production costs.

For **smart cities and IoT-enabled surveillance systems** the study supports the deployment of cost-effective vision solutions for crowd monitoring, traffic analysis and public safety applications. Since such systems often rely on low-cost cameras with limited imaging quality and robustness to noise becomes a critical factor in maintaining reliable performance at scale.

In **unmanned aerial vehicles (UAVs) and remote sensing applications** the framework can enhance object detection in aerial imagery used for disaster management, environmental monitoring, agriculture and wildlife tracking. These applications often involve noisy or degraded data due to atmospheric disturbances, motion blur or sensor limitations which makes robustness a key requirement.

Finally in **wireless and satellite communication systems** the study provides insights into detection performance on compressed or corrupted visual data such as satellite imagery, surveillance feeds and body-worn camera footage. This is particularly important for applications involving bandwidth constraints and lossy transmission where maintaining detection accuracy despite data degradation is essential.

Overall the proposed framework serves as a comprehensive benchmark for evaluating and selecting object detection models in noise-affected environments thereby facilitating the development of more reliable, adaptive and real-world-ready computer vision systems.

6.0 Challenges and Limitations

Despite providing meaningful insights, this study encountered several limitations that must be considered while interpreting the results. One of the primary challenges was noise calibration, as defining realistic noise levels that accurately represent real-world conditions is non-trivial. Although the selected noise intensities were sufficient to evaluate model degradation and identify breaking points, they may not fully generalize to all practical scenarios. Another limitation is the relatively small dataset used for evaluation, which restricts the generalizability of the conclusions; a larger and more diverse dataset would provide more statistically reliable results.

Additionally, the evaluation relies on simplified performance metrics derived from detection outputs, and while robustness trends are clearly observed, a more rigorous assessment would require precise ground truth annotations and full mAP calculations across multiple IoU thresholds. The efficiency analysis is also hardware-dependent, meaning that latency measurements may vary across different systems. Furthermore, certain inconsistencies observed in Poisson noise results indicate that model behavior under signal-dependent noise requires more controlled experimentation. Overall, while the study effectively compares relative robustness across models, it should be extended with larger datasets, standardized benchmarks, and additional evaluation metrics for more comprehensive conclusions.

7.0 Conclusion

The experimental evaluation demonstrates that noise has a significant impact on object detection performance with all YOLO models exhibiting noticeable degradation as noise intensity increases. Among the different architectures used **Gaussian noise** [Table 4] was found to be the most disruptive as causing a rapid decline in mAP and leading to near-complete detection failure at higher noise levels. In contrast models showed relatively better tolerance to lower noise intensities which indicates that modern YOLO architectures can handle minor real-world distortions but struggle under severe perturbations. The comparative analysis across Nano (n) and Small (s) variants highlights a clear trade-off between efficiency and robustness. Lightweight models such as **YOLOv10n** and **YOLOv8n** [Table 5] achieved lower inference latency (≈110–120 ms) [figure 16] while maintaining competitive robustness which making them suitable for real-time applications. On the other hand more complex models like **YOLOv9t** and **YOLOv9s** demonstrated higher precision and F1-scores [table 5] which indicates better feature representation but suffered from significantly higher latency which reducing their practicality in time-critical systems.

The robustness metric (RRI) and average performance drop reveal that **YOLOv9s** is the most robust model with the lowest degradation (≈35.28%) while models such as **YOLO11s** and **YOLO26s** exhibit the highest performance drop (>42%) [table 5] which indicates weaker resilience to noise. However when considering the overall balance between accuracy, robustness, and efficiency **YOLOv10n emerges as the best-performing model** achieving the highest final score (≈0.599) [table 5] due to its strong trade-off between low latency and stable performance under noisy conditions. The efficiency vs robustness analysis further confirms that no single model dominates all aspects infact model selection should depend on application requirements—favouring faster models for real-time deployment and more robust models for accuracy-critical environments. Overall this study provides a systematic evaluation framework and highlights the importance of noise-aware model selection which suggests that future work should focus on improving robustness through data augmentation, noise-resistant training strategies and adaptive preprocessing techniques.

8.0 Future Work

While the present study provides a comprehensive evaluation of YOLO-based object detection models under synthetic noise conditions several promising directions can be explored to further enhance robustness and real-world applicability. A primary way involves the incorporation of noise-aware and noise-augmented training strategies wherein models are retrained or fine-tuned using datasets augmented with controlled noise levels of Gaussian, Salt-and-Pepper and Poisson noise. Such an approach would enable the models to learn invariant and noise-resilient feature representations thereby improving generalization under degraded visual conditions.

Another important direction is the integration of adaptive preprocessing and denoising pipelines. Traditional filtering techniques such as median and Gaussian filters can be employed for specific noise types while more advanced approaches also includes deep learning-based denoising autoencoders and diffusion-based restoration models and can be explored to recover high-quality features prior to detection. Additionally end-to-end architectures that jointly perform denoising and detection could be investigated to minimize information loss during preprocessing.

From an architectural perspective future work should extend the analysis beyond YOLO variants to include transformer-based object detectors (e.g., DETR and its variants) as well as hybrid CNN-transformer models. This would provide deeper insights into how different feature extraction mechanisms—convolutional vs. attention-based—respond to noise perturbations. Evaluating larger-scale YOLO models (medium, large, and extra-large variants) could help understand the trade-off between model capacity and robustness.

The evaluation framework can also be enhanced by incorporating more comprehensive performance metrics including precision-recall curves, mAP across multiple IoU thresholds (mAP@0.5:0.95) robustness indices such as Relative Robustness Index (RRI) and energy efficiency metrics for deployment scenarios. Analyzing failure cases and error distributions under different noise regimes would provide valuable diagnostic insights into model behaviour.

To improve real-world relevance future studies should consider experiments on larger and more diverse datasets and include real-world noisy data collected from cameras operating under adverse conditions such as low light, motion blur, weather disturbances and sensor noise. Domain adaptation techniques and cross-dataset validation can further strengthen the generalizability of the findings.

Finally exploring deployment-oriented optimizations such as model pruning, quantization and edge-device benchmarking would be beneficial for integrating robust object detection models into real-time systems including autonomous vehicles, UAVs, and IoT-based surveillance platforms.

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